

CableTract: A Co-Designed Cable-Driven Field Robot for Low-Compaction, Off-Grid Capable Agriculture

A Prototype-Free Feasibility Framework and Analytical Envelope*

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Abstract

Conventional field operations spend most of their energy moving the tractor, not the implement, and pay a soil-compaction penalty for it. We present **CableTract**, a prototype-free feasibility study of a two-module cable-driven field robot: a stationary Main Unit and a lighter helical-pile Anchor tension a cable across a strip while a lightweight ≈ 250 kg carriage does the work, keeping the heavy bodies on the headland. The central design lever is a 10-implement library *co-designed* for the architecture — narrower, slower, shallower and lighter than tractor implements — which closes the energy, anchor and compaction budgets.

An open, reproducible pipeline couples catenary and anchor mechanics, [American Society of Agricultural and Biological Engineers \(ASABE\)](#) D497.7 draft estimates, an hourly solar/wind/battery simulator on six climate sites, a coverage planner, a contact-pressure compaction model, discounted-cash-flow economics with life-cycle CO_2 , and global sensitivity analysis. Five higher-fidelity studies — a nonlinear p - y anchor model, a closed-loop depth-control co-simulation, a cable-safety model, a soil-tool discrete-element model, and a co-design robustness sweep — stress-test the load-bearing assumptions.

Under the reference assumptions the architecture cuts compacted area by $\sim 98\%$ and useful energy $\sim 4\times$ versus an 80 hp diesel tractor, and is favourable in a tractor-*replacement* frame near capex parity; off-grid operation and additive-purchase economics are climate- and scale-dependent. The dominant unresolved risks are *mechanical* — anchor loading, depth control and cable safety — which the studies bound but a prototype must ultimately test: a consistent analytical envelope for a future prototype, not a validated machine.

Keywords: cable-driven robot; soil compaction; implement co-design; off-grid agriculture; techno-economic analysis; feasibility modelling; discrete element method.

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1 Introduction

A modern 80 hp **Four-Wheel Drive (4WD)** utility tractor weighs ≈ 4 t, of which the implement is typically 200–500 kg. Each pass therefore drags ten to twenty times more body mass than tool mass across the field, paying a soil-compaction penalty (mean contact pressure 100–250 kPa, well above the 50 kPa threshold for clay soils) and burning ≈ 12 L/ha of diesel even on light operations. Over a 15-year lifetime on a 25 ha farm this is 4 500 L of diesel and ≈ 12 t CO₂eq.

CableTract attacks the dead-weight problem at the *architectural* level rather than the powertrain level. Two modules sit on opposite headlands of a strip: a **Main Unit** carrying the winch, motor, controller, battery, **Photovoltaic (PV)** array, and small wind turbine; and an **Anchor** resisting the cable reaction with helical screw piles. A lightweight implement carriage rolls along the cable between them. Only the carriage enters the field. The heavy bodies move only at the end of each strip, by one strip width, along the headland. Replacing the heavy traction body with a tensioned cable changes three things at once: (i) only the carriage's mass compacts the soil; (ii)

the energy budget no longer carries a 4 t body; (iii) the Main Unit, being nearly stationary, can carry deployable PV and a small wind turbine more easily than a fully mobile electric tractor.

The natural objection is that this only works if the implements are also redesigned. A conventional 8-row planter or a 3 m disk harrow is built to be towed by a 4 t tractor; it expects high draft and high speed. Towing those *unmodified* implements with a cable robot is the wrong problem. The right problem is to **co-design the carriage and a small library of implements together**: narrow strip widths matched to the cable swath, slower operating speeds that exploit the v^2 drop in soil draft, shallower depths that match what a carriage-mounted tool can resist, and lighter implement frames freed from the hitch loads of a real tractor. This co-design is the main analytical lever of the paper and we return to it explicitly in [Sections 3.4](#) and [5.5](#).

The contribution is not an experimental validation. It is a **prototype-free, fully reproducible feasibility envelope** that asks one question:

Under what (climate $\times$ farm size $\times$ operation type) conditions does a co-designed cable-driven field robot beat a conventional diesel tractor on energy, compaction, [Net Present Value \(NPV\)](#), and life-cycle CO_2 — and where doesn't it?

We answer it with an open analytical pipeline¹ that runs end-to-end in under ten minutes from a clean checkout, ships with all input data bundled (no live [Application Programming Interface \(API\)](#) calls), and regenerates every figure and table in this paper from the seven phase scripts (`run_phase1`–`run_phase7`). [Section 5](#) walks through the model in the order in which the figures are generated; [Section 7](#) reports the results against four explicit claims; and [Section 9](#) closes the envelope with the binding constraints.

Key results (codesigned reference). Each is conditional on the workload and purchase frame stated in the body:

- **Energy:** 889 Wh/decare² (8.89 kWh/ha) delivered electrical (flat-field bound 921 Wh/decare without regen) — about $4\times$ less *useful* (drawbar) work and $\sim 13\times$ less *primary fuel* energy than the ≈ 120 kWh/ha (12 L/ha) an 80 hp diesel tractor burns; most of the larger factor is generic ICE-to-electric conversion, not the architecture.
- **Compaction:** $\sim 98\%$ less compacted *area* and a $\sim 73\times$ lower per-vehicle contact-pressure index, because only the 250 kg carriage enters the field.
- **Off-grid:** 10–14 decares/day median on harvested energy across the six sites; *climate-conditional* (achievable in solar-rich climates, grid-backed in northern temperate ones), and annual energy-positivity is a weaker claim than hourly autonomy.
- **Economics (replacement frame, near capex parity):** NPV vs diesel 3 575 EUR at 25 ha/yr/8%, positive across 1–100 ha/yr; discounted payback 1.3 yr at 25 ha, below one year by 50 ha. In the *additive* frame the 25 ha NPV is negative (≈ -31 kEUR), positive only above ~ 240 ha/yr.
- **Life-cycle CO_2 :** 14.6 kg/(ha yr) versus 32.5 (diesel) and 22.9 (electric tractor) — a $2.2\times$ improvement independent of grid decarbonisation.

2 Related Work

2.1 Prior cable-driven agriculture

Cable-driven field machines are not new. The closest published prior art is **US 8,763,714 B2** [[1](#)] (Orlando & Zoffoli), which describes a cable traction system using two technically *identical* electric machines, each with a hoist and stabilisation, with the implement moving between them as cable

¹<https://github.com/yilmazozgur/cabletract>

²One decare = 0.1 ha = 1000 m²; 1 Wh/decare = 10 Wh/ha.

is wound and unwound. The patent motivates the design through reduced soil degradation and electric propulsion, and includes a telescopic arm for cable placement during repositioning.

CableTract overlaps with that family but differs in three structural ways:

1. **Asymmetric architecture.** CableTract uses a heavy [Main Unit \(winch + motor + battery + harvester module\) \(MU\)](#) ([Permanent-Magnet Synchronous Motor \(PMSM\)](#), controller, battery, harvester) and a much lighter Anchor (helical pile drive, sensors). The Anchor needs no winch and no high-power electronics, which roughly halves its [Bill of Materials \(BoM\)](#).
2. **Helical-pile anchoring.** Cable reaction is resisted by screw augers driven into the soil rather than by the machine’s mass alone. This is what makes the lighter Anchor module geotechnically defensible ([Section 5.4.4](#)).
3. **Co-designed implements.** CableTract is paired with an implement library sized for the cable architecture, not borrowed from tractor inventories ([Section 3.4](#)).

We do *not* claim CableTract is the first cable-driven agricultural machine. The contribution is an analytical evaluation of the asymmetric [MU](#) + Anchor architecture under co-designed implements, and the identification of the operating envelope in which it is genuinely competitive.

2.2 Adjacent vehicle classes

CableTract sits in a design space already populated by five distinct vehicle classes, and a fair comparison must place the architecture against each of them rather than against diesel alone. We bundle a competitor table³ with public-domain reference numbers for each class ([Section 6](#)). The five classes are:

- **Conventional 80 hp 4WD diesel utility tractor** — cost-floor reference; the thing CableTract has to beat on energy, compaction, and [NPV](#).
- **Battery-electric tractors** (Monarch MK-V, Kubota LXXe-261, Solectrac eUtility) — same form factor as a tractor but with a Li-ion pack. Solves the fuel cost but not the compaction problem; usually grid-charged.
- **Autonomous PV-powered field robots** (Farmdroid FD-20) — small, light, on-board solar; designed for sugar beet / row crops. Cannot do tillage. Off-grid by construction.
- **Autonomous wheeled spot sprayers** (EcoRobotix ARA) — narrow, light, on-board PV; spraying only.
- **Autonomous weeders** (Naio Oz / Orio) — vegetable / row-crop only; no heavy draft.

Across this landscape CableTract is the only architecture that simultaneously (a) is *off-grid capable*, (b) supports *primary tillage* (within its co-designed envelope), and (c) does so with a 250 kg in-field footprint rather than a 1–4 t tractor body. The numerical comparison is in [Section 6](#).

3 System Concept and Co-Design

[Figure 1](#) shows the CableTract concept in operation: the two heavy modules rest on opposite headlands while only a tensioned cable and a lightweight implement carriage enter the field.

³See `tables/competitor_comparison.csv` in the repository.



Figure 1: CableTract operating in a field. The Main Unit (left) houses the winch, drivetrain, battery, PV panel and small wind turbine; the Anchor (right) is a passive helical-pile module that resists the cable reaction. A lightweight implement carriage runs along the tensioned cable across a typical 50 m span. Only the carriage and the cable enter the field — the two heavy modules stay on opposite headlands for the entire operation.

3.1 Two-module architecture

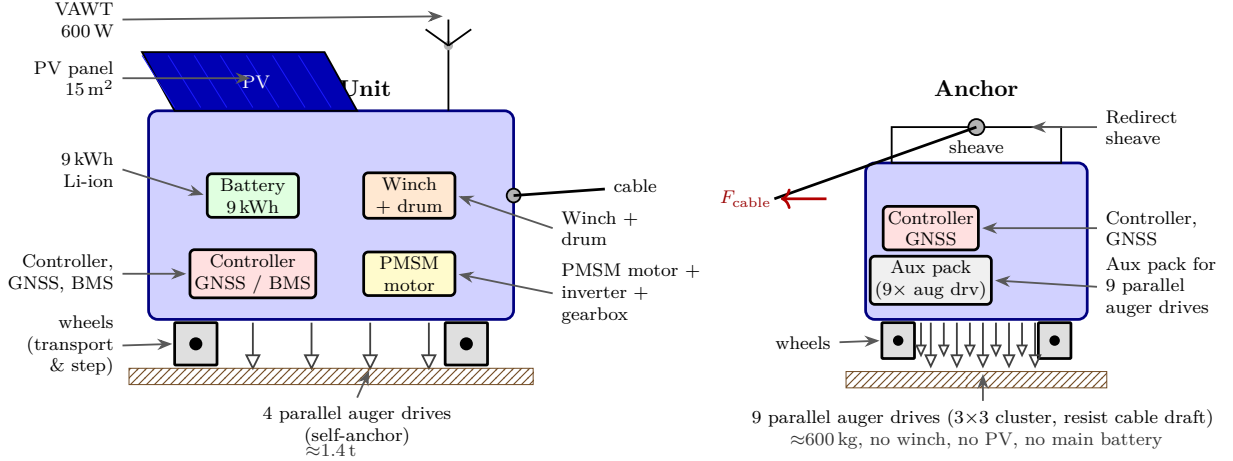


Figure 2: The two physically separable modules of CableTract. **Left:** the **Main Unit** carries the entire active drivetrain (PMSM + winch + drum), the energy stack (9 kWh battery, 15 m² PV, 600 W VAWT) and the controller, and self-anchors with four parallel auger drives during operation (one BLDC drive motor per auger, so all four insert and retract concurrently). **Right:** the **Anchor** is a passive sheave block on top of nine parallel auger drives in a 3×3 cluster that together resist the full horizontal cable reaction F_{cable} at the opposite headland; it carries only an auxiliary pack powering the nine drives and a controller/GNSS for headland repositioning, no winch, no PV, no main battery. The parallel-drive choice on both modules is what keeps the per-round insert/retract cycle short enough to hit the 60 s setup target of Table 3; a single-drive-head design that visited each auger sequentially would not. The asymmetric split halves the BoM and mass of the second module relative to a symmetric two-winch design.

CableTract has two physically separable modules (Fig. 2; the deployed system is sketched in Fig. 1):

- **Main Unit** (≈ 1.4 t including PV deployable): PMSM motor + 3-phase inverter + 2-stage gearbox + 8 mm Dyneema-on-drum winch, 9 kWh Li-ion pack, 15 m^2 flat-plate mono-Si PV, 600 W small Vertical-Axis Wind Turbine (VAWT), controller, Global Navigation Satellite System (GNSS), Battery Management System (BMS), and 4 parallel auger drives for self-anchoring of the Main Unit itself (one small Brushless DC motor (BLDC) per auger, all four cycling concurrently).
- **Anchor** (≈ 600 kg): 9 parallel auger drives in a 3×3 cluster (one BLDC per auger, with a shared auxiliary battery pack and a single controller multiplexing across the nine drivers, so all nine insert and retract concurrently in a single per-round cycle), redirect sheave block, simple electric drive for headland repositioning on its own wheels, GNSS, no main battery beyond the auxiliary auger-drive pack.

In *transport mode* the Anchor docks against the MU and the assembly behaves as a compact 1.5 m wide, 1 m tall electric utility vehicle that can drive between fields on farm tracks. In *work mode* the Anchor walks across the field on its own electric drive and plants itself at the far headland; the MU runs the cable across the strip; the implement carriage rolls along the cable.

3.2 Operating cycle

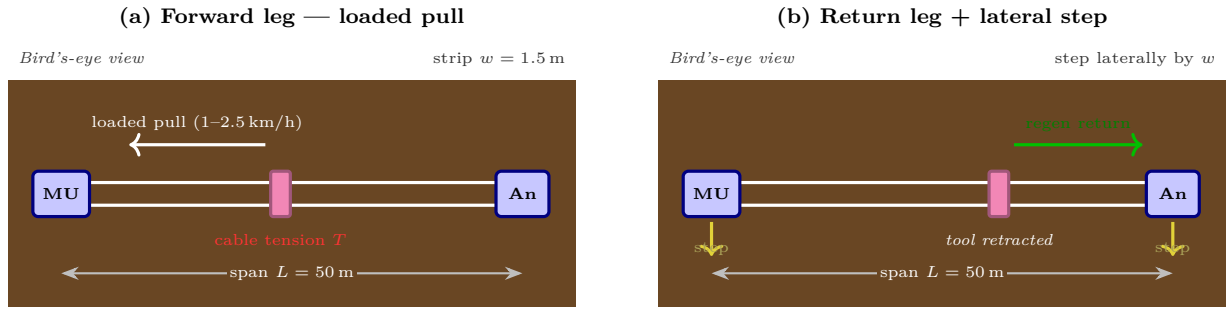


Figure 3: One round of the CableTract operating cycle, bird's-eye view. **(a) Forward leg:** the Main Unit's winch reels in cable, the implement carriage moves loaded toward the MU at 1–2.5 km/h, the working tool engages the soil, and the full draft F_{draft} is reacted at the Anchor as F_{reaction} . **(b) Return leg:** the tool retracts, the cable goes slack, and the carriage rolls back to the Anchor end (downhill assist or active pull-back). The unloaded return leg recovers kinetic / potential energy via four-quadrant motor operation by default (Sections 5.4.6 and 5.12.4). After the return, the MU and the Anchor each step laterally by one strip width (1.5 m) along the headland and the cycle repeats on the next strip.

One round of work proceeds as (Fig. 3):

1. The implement carriage starts at the Anchor end with the working tool engaged.
2. The MU's winch reels in cable. The carriage moves loaded toward the MU at the operating speed (typically 1–2.5 km/h).
3. At the MU end, the carriage tool retracts, the cable goes slack, and the carriage either gravity-rolls back (downhill) or is towed back by a thin return line. The unloaded return leg can recover kinetic / potential energy via four-quadrant motor operation (regenerative braking, Section 5.4.6).
4. The MU and Anchor each step laterally by one strip width along the headland.
5. Setup repeats on the next strip.

Typical numbers for the codesigned reference: span 50 m, strip width 1.5 m, carriage mass ≈ 250 kg, operating speed 1.5 km/h, setup time per round 60 s.

3.3 Why an asymmetric architecture

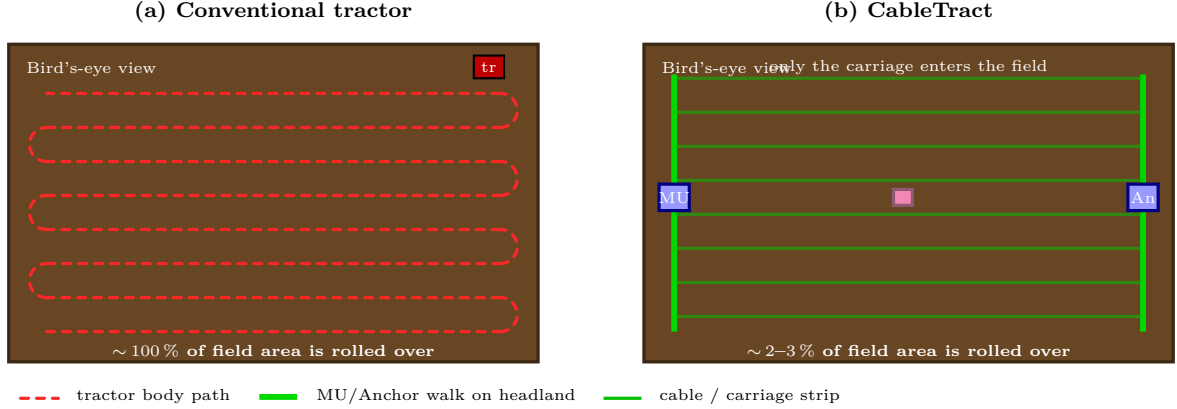


Figure 4: Why an asymmetric, headland-only architecture wins on compaction. **(a)** A conventional tractor must drive a boustrophedon (snake) path that covers essentially the entire field area at least once per pass — every square metre of the soil is rolled over by a 1–4 t body. **(b)** CableTract keeps the Main Unit and the Anchor on opposite headlands and runs only a 250 kg carriage along the cable across the field; over a full season the rolled-over fraction collapses to $\sim 2\text{--}3\%$ of the field area (essentially the carriage strips). The contact-pressure $\times$ pass-count integrand drops by a much larger factor still — see Fig. 16 and Section 7.1 for the quantitative compaction model.

Figure 4 contrasts CableTract’s headland-only coverage with a tractor’s full-field boustrophedon path. Symmetric cable-traction systems (like the Orlando–Zoffoli patent [1]) need two winches, two batteries, two motor controllers, and two sets of high-power electronics. The asymmetric CableTract design moves all the power electronics into one module and lets the second module be a simple anchored drive. This roughly halves the BoM of the second module, halves its mass, and removes one of the two failure-prone winches.

The price of the asymmetry is that the cable reaction is one-sided. The Anchor must hold the entire horizontal draft against the soil. We address this with helical screw piles (Section 5.4.4) and quantify the per-auger requirement against *two* published lateral-capacity references — a realistic medium-dense fixed-head nominal and a deliberately pessimistic loose-sand bound — so that every auger count is reported with its soil assumption made explicit.

3.4 Co-designed implement library

The single most impactful design choice in this paper is **building implements for CableTract instead of borrowing them from a tractor**. A conventional 8-row planter is built to survive the hitch loads of a 4 t tractor at 9 km/h on a 1.5 m depth-control wheel; it weighs 800 kg and pulls 3.9 kN of draft. CableTract neither needs nor can usefully exploit any of those numbers. The codesigned 4-row planter⁴ is sized for a 1.5 m strip, 2 km/h operating speed, $0.85\times$ the conventional working depth, and a carriage frame freed from tractor hitch loads. Its **50th (median) percentile of a sampled distribution (P50)** draft drops from 3.88 kN to 1.94 kN — a **factor of 2.0 reduction** (an 8-row tool becomes a 4-row tool on the narrow strip, run slower). We are careful below to separate which part of this is genuine implement re-engineering and which is simply a lighter operating point.

The 10-implement codesigned library and the achieved draft reduction (Table 1):

⁴Tabulated in `data/asabe_d497_cabletract.csv` of the repository.

Table 1: Codesigned implement library, P50 draft reduction relative to the conventional American Society of Agricultural and Biological Engineers (ASABE) D497.7 [2] entry it derives from. Medium-texture soil, samples from Section 5.5.2.

Operation	Conventional implement	Conv. P50 (N)	Codesigned implement	Codes. P50 (N)	Reduction
primary tillage	subsoiler 1-shank	10 981	narrow ripper 1-shank (15 cm)	3 022	0.28×
primary tillage	chisel plow 7-tool	14 174	narrow chisel 4-tool (12 cm)	4 219	0.30×
secondary tillage	disk harrow 2.5 m tandem	8 428	narrow disk 1.5 m	3 231	0.38×
secondary tillage	field cultivator 11 S-tine	3 175	narrow cultivator 6-tool	1 347	0.42×
primary tillage	sweep plow 3 m	15 412	narrow sweep 1.5 m	5 174	0.34×
seeding	row planter 8-row	3 876	codesigned planter 4-row	1 935	0.50×
seeding	grain drill 24-row	9 302	codesigned drill 12-row	4 650	0.50×
weeding	rotary hoe 4 m	2 251	codesigned rotary hoe 2 m	788	0.35×
spraying	boom sprayer	204	codesigned sprayer 1 m	143	0.70×
mowing	rotary mower 3 m	1 452	codesigned mower 1.5 m	523	0.36×
Median reduction					0.37×

Median reduction across the 10 operations: $0.37\times$ ($\approx 63\%$ lower draft). The reduction comes from three independent levers, each defensible from the ASABE D497.7 equation

$$D = F_i (A + Bv + Cv^2) WT, \quad (1)$$

where D is draft (N), v is field speed (km/h), W is the implement’s natural width unit, T is working depth, F_i is a soil-texture multiplier, and (A, B, C) are machine-specific coefficients. The three levers are:

1. **Width.** Halving W halves draft directly. The cable architecture wants narrow strips because the 50 m cable span and the 1.5 m strip width together produce a 75 m^2 working envelope per round, which matches the carriage’s mechanical envelope.
2. **Speed.** The $Bv + Cv^2$ terms grow rapidly with speed. Operating at 1.5–2.5 km/h instead of 5–9 km/h drops the speed-dependent draft contribution by roughly 60%. This is a *physical* gain, not a marketing claim, and it is the same lever any slow-moving robot exploits. Figure 9 quantifies the saving.
3. **Depth.** Working at $0.85\times$ the conventional depth drops T by 15% directly. CableTract takes one extra pass per season for primary tillage to compensate, which is bookkept in the energy and economics models.

Accounting of the levers. Two of these three levers are *operating-point* choices rather than implement re-engineering, and we state this plainly. For the depth-dependent tillage tools (narrow ripper, chisel, disk, sweep, cultivator) the codesigned entries in `data/asabe_d497_cabletract.csv` carry the *same* D497.7 (A, B, C) coefficients as the conventional implement they derive from; the draft reduction is obtained by running the same tool geometry shallower, slower, and on a narrower strip. This is physically real — Eq. (1) is linear in W and T and rises with v — but it carries an agronomic obligation: a shallow narrow pass is not interchangeable with a deep wide one for every tillage objective (a 15 cm ripper pass does not fracture a deep plough pan the way a 40 cm subsoiler pass does), which is why we bookkeep an extra primary-tillage pass and restrict the heaviest operations to favourable soil in Section 5.4.4. The row-count halving for the planter and drill (8→4 rows, 24→12 rows) halves per-pass draft but also halves the area covered per pass, so it is throughput- and energy-per-area-neutral; its benefit is the lighter frame and slower speed, not the row count itself. Only the lighter tools (rotary hoe, mower, sprayer) gain a genuine reduction in the constant term A from a frame freed of tractor hitch loads. We therefore present the $0.37\times$ median as the achieved draft ratio at the codesigned *operating point*, not as a claim that the tools are intrinsically half as draft-hungry. Table 2 states, per operation, where the reduction comes from and whether the codesigned pass is agronomically equivalent — a design judgement to be confirmed by a prototype, not a measured result.

Table 2: Agronomic-equivalence status of each co-designed operation. “Source” separates genuine implement re-engineering (a lighter frame lowers the constant term A) from operating-point choices (narrower W , slower v , shallower T , with D497 coefficients unchanged). Equivalence is a design judgement for a prototype to test.

Operation (conventional → codesigned)	Draft-reduction source	Agronomic-equivalence status
subsoiler → narrow ripper (15 cm)	depth + speed (coeffs unchanged)	<i>Not</i> equivalent for deep pan; needs 2 passes / favourable soil
chisel 7-tool → narrow chisel 4-tool	depth + speed + tool count	Approx. equivalent with an extra pass
sweep 3 m → narrow sweep 1.5 m	width + depth	Approx. equivalent (more strips)
disk 2.5 m → narrow disk 1.5 m	width + lighter frame	Likely equivalent (secondary tillage)
cultivator 11-tine → 6-tool	width + tool count	Likely equivalent
planter 8-row → 4-row	row count (area-neutral)	Equivalent (twice the strips)
grain drill 24-row → 12-row	row count (area-neutral)	Equivalent (twice the strips)
rotary hoe 4 m → 2 m	width	Equivalent
boom sprayer → 1 m sprayer	width + lighter frame (A)	Equivalent
rotary mower 3 m → 1.5 m	width + lighter frame (A)	Equivalent

The price of the codesigned library is that the farmer cannot take a CableTract to a neighbour’s field and hook up the neighbour’s plough. CableTract is sold as a *system* — Main Unit + Anchor + carriage + implements — not as a generic cable robot that accepts off-the-shelf tools. This is the same business model as Farmdroid, EcoRobotix, and Naio.

4 Research Questions

We organise the analysis around four claims and seven research questions. The four claims are the headline assertions the paper defends with code-traceable numbers; the seven research questions decompose each claim into the operational regime, the physical mechanism, and the binding constraint.

C1 — Compaction. CableTract reduces in-field compacted area by at least 90% and the contact-energy index by at least $20\times$ compared with an 80 hp 4WD diesel tractor on the same field, because only the lightweight carriage rolls inside the field while the MU and Anchor stay on the headland.

C2 — Energy. With a co-designed implement library and regenerative braking on the return leg (the default design), CableTract operates at ≈ 0.89 kWh/decare (≈ 8.9 kWh/ha) of delivered electrical energy — about $4\times$ less *useful* (drawbar) work, and $\sim 13\times$ less *primary fuel* energy, than a diesel tractor on the same operation, driven by (i) the absence of dead-weight transport, (ii) the v^2 speed reduction, (iii) the narrow-strip width, (iv) a small return-leg regenerative recovery, and (v) electric rather than thermal conversion.

C3 — Off-grid. Off-grid operation is achievable in solar-rich climates (Konya, Ludhiana, São Paulo) with 15 m^2 of PV and a 9 kWh battery, conditional in Mediterranean ones (Palencia, Des Moines), and not achievable at any practical hardware scale in northern temperate climates (Beauce) under a 6 h/day duty cycle. The off-grid claim is *climate-conditional*, not categorical.

C4 — Economics and life-cycle CO₂. The codesigned CableTract is NPV-positive (replacement frame) across the full 1–100 ha farm-size range at all three discount rates we test (5/8/12%), with discounted payback of 1.30 yr at 25 ha annual operating area and below one year by 50 ha, and lifecycle CO₂ of 14.6 kg/(ha yr) versus 32.5 for diesel — a $2.2\times$ improvement that does not depend on grid decarbonisation.

The research questions:

Research Question (RQ)1

What is the compacted-area reduction and contact-energy reduction across rectangle, L-shape, and irregular-concave field classes?

RQ2

What energy per decare does the codesigned reference achieve, and how does it decompose between mechanical work, drivetrain losses, and dead-weight motion?

RQ3

Under which combinations of annual [Global Horizontal Irradiance \(GHI\)](#), panel area, and battery capacity is the system off-grid capable?

RQ4

For which operation classes is the architecture most favourable, and where does the helical-pile anchor envelope bind?

RQ5

How do polygon shape and size erode the throughput on real fields?

RQ6

Is the codesigned reference Pareto-optimal on $(\text{CAPEX} \times \text{throughput} \times \text{payback} \times \text{energy intensity})$, or does it leave headroom?

RQ7

Across the $(\text{annual GHI} \times \text{farm size})$ plane, where is CableTract NPV-positive against diesel, and where does the off-grid feasibility envelope bind first?

5 Methods

5.1 Modelling philosophy

The analysis is parametric and designed to expose its assumptions. Every parameter has units, every default value has a citation in a bundled data file, and every claim in [Section 7](#) has a per-figure CSV companion with the underlying numbers.⁵ We do not attempt high-fidelity soil-tool dynamics, full closed-loop control simulation, or finite-element cable modelling. We use first-principles work-energy balances, the standard [ASABE D497.7](#) implement-draft equation, an hourly [Typical Meteorological Year \(TMY\)](#)-based PV+battery simulator, a static contact-pressure compaction model, a discounted-cash-flow economics engine, and global Sobol sensitivity over 20 inputs. This is the appropriate fidelity for an early-stage feasibility study.

Every higher-level analysis is composed on top of one atomic computation that takes a parameter set and returns a result record. The energy simulator ([Section 5.7](#)), the global sensitivity decomposition ([Section 5.11.3](#)), the architectural variants ([Section 5.12](#)), and the operating envelope ([Section 5.13](#)) are all callers of this same routine. Every figure is therefore re-derived from a single physics-and-economics chain rather than from parallel approximations that drift apart.

5.2 Codesigned reference parameter set

The codesigned reference is the single system the rest of the paper analyses. It is *the* baseline: there is no other parameter set we report headline numbers on, so every result in [Section 7](#) can be traced back to the values in [Table 3](#), which lists each entry together with a one-line engineering justification.

⁵Full source and data: <https://github.com/yilmazozgur/cabletract>

Table 3: Codesigned reference parameter set. Used unchanged in every figure, table, and headline number in the paper.

Parameter	Value	Notes
span (m)	50	Anchor-to-Main-Unit cable distance
strip width (m)	1.5	Matches the codesigned implement library
implement + cable transport load (N)	600	Carriage tool + cable drag carried along the span (F_i in Eq. (2)); the carriage <i>structural</i> mass is ≈ 250 kg and enters the compaction model (Section 5.10)
draft load (N)	1 800	Representative P50 for the medium-load operational calendar; the full 10-implement library median is 2 479 N (Section 5.5.1), with heavy primary tillage higher
system repositioning load (N)	2 200	Rolling-resistance force as the ≈ 2 t MU + Anchor step one strip width along the headland (F_s in Eq. (2)); a force, not a mass
drivetrain efficiency (effective)	0.518	One-way chain 0.50 (6 components, Section 5.4.1) + $\approx 3.5\%$ four-quadrant recovery (slope PE + KE; ≈ 0 on flat)
PV area (m ²)	15	Mono-Si flat plate, deployable
wind generator (W)	100	Typical harvested power; a 600 W-rated helix VAWT ($C_p = 0.30$) that reaches nameplate only near 12 m/s (Section 5.7.3)
setup time per round (s)	60	Anchor placement + cable spool + carriage hookup
battery (kWh)	9	Li-ion, 96/96 % charge/discharge eff.
operating window (h/day)	10	Daily total work window
operating days/yr	170	Standard EU agronomic calendar
diesel reference fuel (L/decare)	1.2	Mid-range field operation (12 L/ha) [10, 2]
diesel price (EUR/L)	1.40	EU 2024 agricultural diesel
capex CableTract (EUR)	35 870	Itemised in Section 5.8
project horizon (yr)	15	Matches Kubota / Monarch warranty bands
discount rate (%)	8	EU farm equipment loan rate

5.3 Pass model

For one round of work, useful mechanical work is

$$E_{\text{mech,round}} = F_d L + F_i L + F_s W, \quad (2)$$

where F_d is the implement draft (N), F_i the carriage-plus-cable transport load (N), F_s the system-repositioning force (N), L the cable span (m), and W the lateral repositioning distance per round (m). For the codesigned reference the breakdown is $F_d L = 90$ kJ (72 %), $F_i L = 30$ kJ (24 %), and $F_s W = 4.4$ kJ (4 %): the in-field work ($F_d L + F_i L$, 96 %) dwarfs the headland-repositioning work ($F_s W$, 4 %), which is exactly the dead-weight saving relative to a tractor (where the system-repositioning term, paid over the whole field rather than once per strip, dominates). The $F_i L$ term charges the full transport load over the entire span, which slightly *over*-estimates CableTract’s energy (a rolling carriage does no net lifting work over a level span), so the 889 Wh/decare headline is conservative in this respect. (These fractions are of the round-trip mechanical work; the headline energy then divides by the regen-default effective drivetrain efficiency of Section 5.4.1.)

5.4 Cable mechanics, drivetrain, anchor

The cable mechanics and drivetrain model are verified against 22 analytic invariants (sag monotonicity, Hooke’s law in the elastic regime, energy conservation on the regen leg, and others).⁶

5.4.1 Decomposed drivetrain efficiency

A lumped winch efficiency $\eta_w \approx 0.5$ is the most common single-parameter shortcut in agricultural-electrification work. It collapses six independent loss mechanisms — motor copper losses, inverter

⁶Test suite: `tests/test_physics.py` in the repository.

switching losses, gearbox friction, drum tension losses, pulley redirect losses, cable elongation hysteresis — into one number, which a global sensitivity analysis (Section 5.11.3) would then assign almost all of the variance to. The chain we use in this paper is the explicit product of those six components, each independently sourced from a datasheet or a published engineering standard:

$$\eta_{\text{drivetrain}} = \eta_{\text{motor}} \eta_{\text{ctrl}} \eta_{\text{gearbox}} \eta_{\text{drum}} \eta_{\text{pulley}} \eta_{\text{cable}}. \quad (3)$$

Two presets are bundled (Table 4); the “baseline” *one-way* product matches the lumped 0.5 by construction, and “premium” recovers $\approx 74\%$. The global sensitivity analysis in Section 5.11.3 then samples each component independently, so the Sobol decomposition sees six separate variance contributors instead of one undefended fudge factor.

This one-way chain governs *motor sizing* (the peak loaded pull, Section 5.4.5). The *energy budget* uses a slightly higher effective efficiency, because the default four-quadrant drive recovers a *modest* amount of energy on the unloaded return leg: gravitational potential energy on sloped fields (Eq. (8)), the carriage’s kinetic energy at each stroke end, and dissipation a one-quadrant drive would otherwise lose to braking. We are deliberately conservative about its size. The draft work itself — the dominant ~ 90 kJ per pass — is dissipated irreversibly in cutting soil and *cannot* be regenerated; the carriage kinetic energy at 2 km/h is only ~ 40 J; and on a perfectly flat field the gravity term of Eq. (8) is zero. We therefore credit only $\approx 3.5\%$ (slope-averaged over the gently-undulating fields of the target regions), lifting the effective drivetrain efficiency from the one-way 0.50 to ≈ 0.518 — *not* the large “half the round is unloaded” recovery a naive argument would suggest. This gives a headline 889 Wh/decare, against a conservative flat-field bound of 921 Wh/decare (the unidirectional drivetrain, Section 5.12.4); the slope-dependent upside is parameterized in Section 5.4.6. The four-quadrant drive is a modest hardware premium (≈ 300 EUR, folded into the Main Unit) and is standard equipment primarily for smooth torque control and slope recovery, not as a large flat-field energy lever.

Table 4: Decomposed drivetrain efficiency. Baseline product reproduces the lumped 0.5; premium recovers 0.74.

Component	Baseline (low-cost)	Premium (best-of-class)	Source
PMSM motor	0.85	0.93	[3]
Three-phase inverter	0.92	0.96	[3]
2-stage planetary gearbox	0.88	0.95	drive-vendor data
Drum capstan + bending	0.88	0.95	[8]
Redirect sheave block	0.90	0.95	[8]
Cable cyclic loss	0.92	0.97	Marlow Excel D12 datasheet
Product	0.50	0.74	—

5.4.2 Catenary sag

The first geometric requirement for any cable-based field robot is that the cable must not sag into the soil. If midspan sag exceeds the implement-attachment height, the geometry breaks before any draft, energy, or economics question can be asked. Figure 5 resolves this with a closed-form bound: it computes the minimum horizontal tension required to keep midspan sag below the depth-control clearance budget, as a function of cable material, span, and self-weight.

A perfectly flexible cable of self-weight w (N/m) hung between two supports a horizontal distance L apart at horizontal tension T_h takes the catenary shape [7]

$$y(x) = a(\cosh(x/a) - 1), \quad a = T_h/w, \quad (4)$$

with midspan sag

$$s = a \left(\cosh\left(\frac{L}{2a}\right) - 1 \right). \quad (5)$$

For $wL/(2T_h) < 0.3$ the parabolic limit

$$s \approx \frac{wL^2}{8T_h} \quad (6)$$

is accurate to 1 %. Both forms are computed and cross-checked at every operating point.

Figure 5 sweeps horizontal cable tension from 0.5 kN to 25 kN for steel 6×19 Independent Wire Rope Core (IWRC), Dyneema SK78 12-strand, and Spectra 1000 fibre rope (all 8 mm nominal) at three spans $L \in \{50, 75, 100\}$ m.⁷

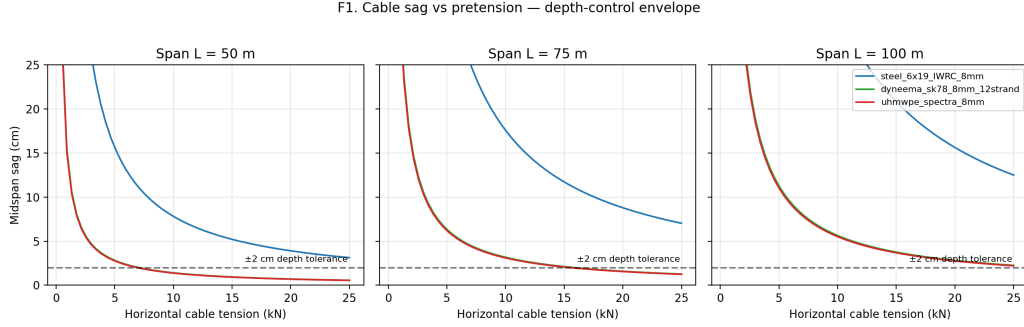


Figure 5: Catenary sag versus horizontal cable tension for three rope materials at three spans. *Tilling depth is set by the carriage’s own depth wheel, not by cable sag* (Section 5.4.3); the cable’s only geometric job is to stay clear of the soil/crop. For a cable support height of ≈ 0.8 m and a ground-clearance budget of ~ 0.3 m (a sag limit of ~ 0.5 m), 8 mm Dyneema (self-weight 0.046 kg/m) needs only ≈ 0.3 kN pretension at the 50 m reference span and ≈ 1.1 kN at 100 m (parabolic limit $T \approx wL^2/8s$). This modest sag-floor pretension is below the implement draft for all but the lightest operations, so the Anchor reaction is governed by the *draft* (Section 5.4.4), not by sag. (A much tighter ± 2 cm tolerance, were it required of the cable, would demand ≈ 7 kN at 50 m — but that tolerance belongs to the carriage’s depth control, not to the cable.) Steel needs $\sim 5\times$ more tension for the same clearance because of its $\sim 5\times$ higher self-weight, which is why CableTract uses synthetic rope.

What this gives the rest of the paper. Figure 5 narrows the design choice to synthetic rope and quantifies the minimum working tension as a function of span. CableTract uses 8 mm Dyneema throughout. The minimum-tension number flows into Section 5.4.3 (which side of the draft-bound vs sag-bound regime are we in?) and into Section 5.4.4 (the helical-pile capacity must absorb the horizontal tension we just selected).

5.4.3 Tension balance and operating regime

The implement carriage’s own depth-control mechanism — not the cable — sets tilling depth. The cable transmits the horizontal draft F_d and stays clear of the soil. Two regimes arise:

- **Draft-bound:** F_d is large enough that the resulting horizontal tension keeps midspan sag below the clearance budget $h_p - c_{\min}$. Horizontal tension equals F_d , anchor sees $\approx F_d$.
- **Sag-bound:** F_d is too small to keep the cable taut against its own weight. The Main Unit over-tensions to $T_{\text{sag},\min}$. The anchor sees this larger tension.

The tension-balance solver detects which regime applies at each operating point and reports the tensions seen by both ends of the cable. Figure 6 sweeps draft from 200 N to 4 000 N at 8 mm Dyneema, span 50 m, $h_p = 1.5$ m, with the per-auger Anchor capacity bands overlaid.

5.4.4 Anchor reaction envelope

The asymmetric architecture of Section 3.3 only works if the lighter Anchor module can geotechnically resist the cable reaction. A mass-anchored solution would force the Anchor to weigh as

⁷Mechanical properties from manufacturer datasheets, tabulated in `data/cable_props.csv`.

much as a small tractor, which would defeat the architecture’s selling point. We therefore replace mass anchoring with helical screw piles and quantify how many augers are required at each draft level. This is the load-bearing question behind the “CableTract is mechanically possible” claim.

The literature on lateral capacity of helical piles spans nearly two orders of magnitude depending on soil density, shaft fixity, embedment, and the chosen serviceability limit, so we anchor the analysis between a realistic nominal and a deliberately pessimistic bound rather than picking one number. The **Magnum Piering 2024 design tables** [5] report allowable lateral capacities of 14–20 kN per small fixed-head helical pile in medium-dense sand (SPT $N \approx 10\text{--}30$) at the IBC2021 1 inch deflection limit, and the industry-standard CHANCE Technical Design Manual [6] brackets the same range. As our *nominal* design basis we adopt a heavily conservative downscale of this — 2 kN per auger (a 7–10 $\times$ reduction on the reported fixed-head values), at a working safety factor of 1.15 — which still corresponds to the medium-dense, fixed-head conditions a real installer would target. As a *worst-case stress bound* we additionally carry 400 N per auger, our per-pile reading of the 4-pile group raft tests of **Khand et al. (2024)** [4] for short piles in *loose* sand at strict deflection thresholds with free-head boundary conditions; we flag explicitly that this number is extracted from a model-scale experiment, so it is used only as a pessimistic ordering bound, not as a field-calibrated capacity. Reporting both makes the soil assumption behind every auger count explicit. The codesigned 9-auger Anchor is sized so that even at the 400 N worst-case bound it anchors the codesigned reference draft (P90 3.0 kN, 9 augers); at the 2 kN nominal those same 9 augers carry the entire codesigned library — and most conventional implements — with a 4–5 $\times$ margin on auger count. We do not lean on the worst case to make the design work; we use it to show how much pessimism the design tolerates. The 9-auger capacity lines plotted in [Figs. 6 and 8](#) are *nominal* (9 $\times$ per-auger, i.e. 3.6 kN on the Khand bound and 18 kN on the Magnum nominal); the auger *counts* we quote already apply the 1.15 working safety factor, $n = \lceil 1.15 T/c \rceil$, so they are allowable-capacity counts, not nominal.

medium-dense soil regime, in which it carries the entire codesigned library with comfortable margin and is competitive even with conventional implements. In the worst-case loose-sand regime, the same Anchor is restricted to the lighter half of the codesigned library (cultivator, planter, rotary hoe, sprayer, mower, and the narrow ripper at its P50). The codesigned library is therefore not a worst-case geotechnical fix — it is the design choice that simultaneously makes the energy budget work, the motor sizing work, and the medium-dense-soil anchor envelope work for the entire operation calendar. In loose-sand sites, the operating envelope shrinks to the lighter half of the library, and heavy primary tillage either waits for soil moisture conditions that improve fixity or is performed by a conventional implement on the same site.

The two-reference bracket above is deliberately agnostic about which per-auger capacity is “correct”. [Section 8.2](#) closes that gap with a nonlinear p - y Winkler model that derives a single internally-consistent per-auger working nominal of 1.6–2.3 kN — which brackets the Magnum nominal and sits above the conservative Khand floor — implying only ≈ 3 augers at the codesigned P90. We nevertheless retain the installed 9-auger count: the p - y result simply reframes it as a deliberate $\approx 3\times$ margin and as redundancy, at no throughput cost (the augers are set once, in parallel, at strip setup), and the worst-case-bound counts quoted above stand as the conservative envelope.

5.4.5 Peak vs continuous motor power

Day-averaged power numbers are convenient for energy budgets but misleading for motor sizing. A real motor must absorb the peak load — start-up, loaded acceleration, transient draft spikes from the implement entering harder soil — and the BOM cost of the motor, inverter, and cooling sits on that peak rather than on the average. We therefore separate the two.

Motor sizing is governed by the peak load, not the day-averaged power. We separate the two as

$$P_{\text{cont}} = \frac{F_{\text{cont}} v_{\text{cont}}}{\eta_{\text{drivetrain}}}, \quad P_{\text{peak}} = \frac{F_{\text{peak}} v_{\text{peak}}}{\eta_{\text{drivetrain}}}. \quad (7)$$

For the codesigned reference at 1.8 kN P50 / 3.0 kN P90 ([Table 5](#)), the codesigned reference uses a **2.0 kW continuous / 3.0 kW peak PMSM**. This is roughly half the motor that an off-the-shelf-implement CableTract would need, and it shrinks the inverter, the gearbox, and the heat-sink mass in proportion. The cost saving propagates into [Section 5.8](#).

Table 5: Peak vs continuous motor power for the codesigned reference at the two drivetrain efficiency presets.

Drivetrain	Continuous (kW)	Peak (kW)
Baseline ($\eta = 0.50$)	1.50	2.50
Premium ($\eta = 0.74$)	1.01	1.69

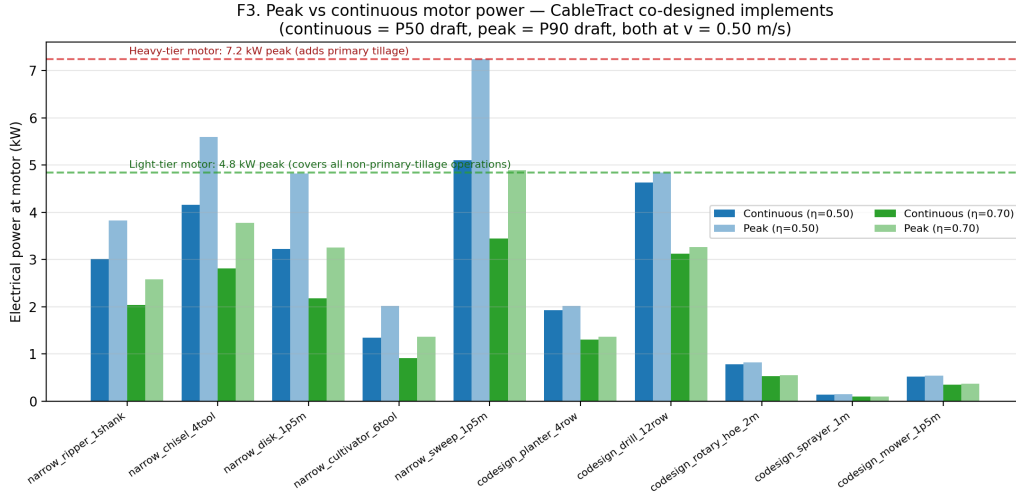


Figure 7: Peak versus continuous motor power across the codesigned implement library at the two drivetrain efficiency presets.

5.4.6 Regenerative return leg

A genuine energy advantage of the cable architecture is that the unloaded return leg lifts no implement, so any potential-energy budget (downhill slopes, carriage descent at end of row, elastic recoil of the cable as load is removed) can be recovered with four-quadrant motor operation. The recoverable energy on a return leg of length d on a slope θ is

$$E_{\text{regen}} = \eta_{\text{regen}}(mg \sin \theta - mg \cos \theta \mu_r) d, \quad (8)$$

with $\eta_{\text{regen}} = 0.55$ and rolling resistance $\mu_r = 0.06$. On flat fields the gravity term vanishes and the formula returns zero, so the dominant regen contribution comes not from slope but from the four-quadrant motor recovering kinetic energy from the cable retraction itself; this is the pathway the default drivetrain exploits (Section 5.4.1), isolated against a no-regen drivetrain in Section 5.12.4 and Fig. 22. Slope-driven regen only adds value on slopes steep enough to overcome rolling resistance ($\sin \theta > \mu_r$), i.e. above $\sim 6\%$ grade.

5.5 Draft model — ASABE D497.7 with the codesigned library

Section 3.4 introduced the codesigned implement library in design terms; the central quantitative claim is that re-sizing implements for the cable architecture cuts P50 draft by $\sim 63\%$. To make that number reproducible rather than anecdotal, we evaluate the standard ASABE D497.7 [2] draft equation as a Monte Carlo over speed, depth, and soil moisture for *both* the conventional and the codesigned libraries on the same axes (Figs. 8 and 9). D497.7 is an experimentally calibrated industry standard [2, 9], so the comparison rests on a published equation rather than on hand-picked single points.

The implement-draft model is the standard ASABE D497.7 [2] equation Eq. (1). F_i is the soil-texture multiplier (Table 3 of D497).

5.5.1 Codesigned implement library

The 10 codesigned implements are tabulated with citations to the conventional D497 [2] entry each derives from and the re-sizing rationale. The library is verified against the canonical D497 example “moldboard plow at 8 km/h, 15 cm depth, fine soil” to better than 1%, and the resulting per-implement P50/P90 magnitudes reproduce the field measurements of Kheiralla et al. [9] on the same texture class to within $\pm 10\%$.⁸

⁸Library and verification: `data/asabe_d497_cabletract.csv` and `tests/test_soil.py`.

5.5.2 Stochastic draft sampler

For each implement we draw 5 000 Monte Carlo samples uniformly over

$$v \in [1, 4] \text{ km/h}, \quad T \in [T^* - 5, T^* + 5] \text{ cm}, \quad \theta \in [0.12, 0.28],$$

with T^* the implement's typical working depth on a medium-texture soil. Figure 8 reports the per-implement P10/P50/P90 draft distribution for the codesigned library, with the two 9-auger Anchor envelopes from Fig. 6 overlaid. The heaviest codesigned operation is the narrow sweep at 5.2 kN P50 (P90 7.25 kN); the lightest non-trivial operation is the rotary hoe at 0.79 kN P50.

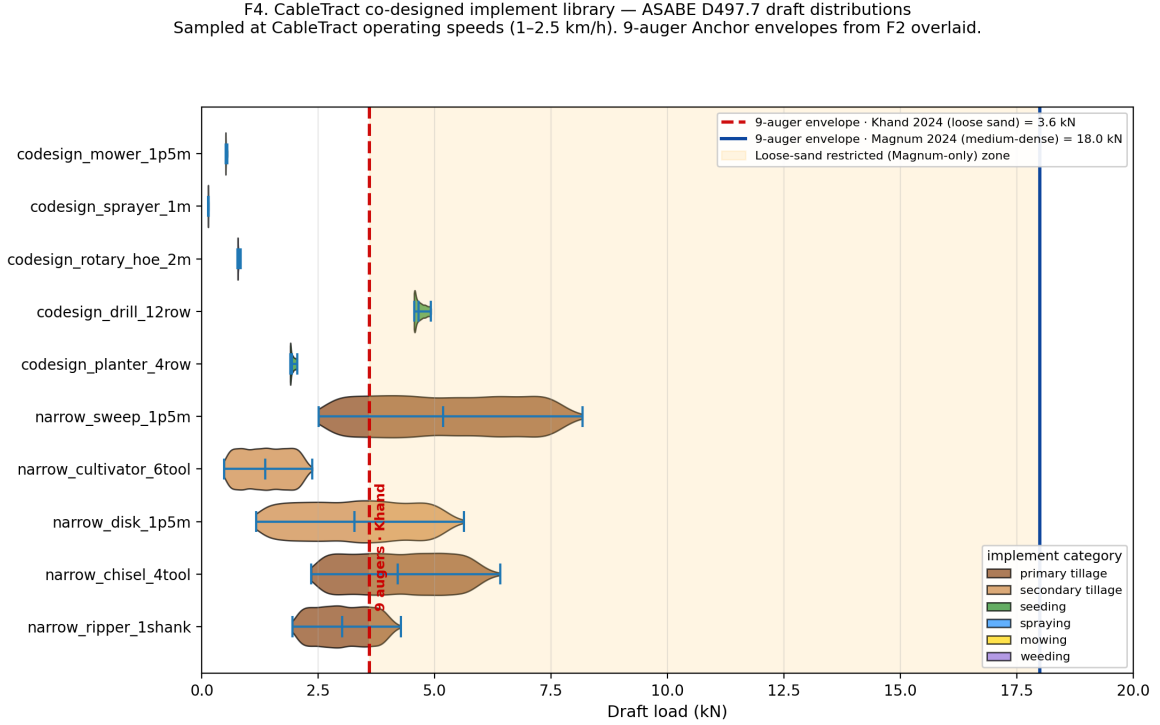


Figure 8: ASABE D497.7 [2] draft distributions for the 10-implement CableTract codesigned library, sampled at CableTract operating speeds (1–2.5 km/h). The two vertical lines are the same 9-auger Anchor envelopes plotted in Fig. 6: the dashed red line at 3.6 kN is the worst-case loose-sand bound (Khand 2024 [4], 400 N per auger); the solid blue line at 18 kN is the realistic medium-dense bound (Magnum 2024 [5], 2 kN per auger). The yellow band between them is the loose-sand-restricted (Magnum-only) zone. Five codesigned implements — mower, sprayer, rotary hoe, planter, and cultivator — sit safely to the left of the Khand line. The narrow ripper sits at the Khand line. Four codesigned implements — narrow sweep, narrow chisel, narrow disk, and codesign drill — fall in the loose-sand-restricted zone and require the medium-dense Magnum reference to fit inside the 9-auger Anchor. The entire library sits well to the left of the Magnum line.

The headline finding: **the entire codesigned library fits inside the medium-dense 9-auger Anchor envelope (18 kN)**, and the lighter half also fits inside the worst-case loose-sand envelope (3.6 kN). The heaviest codesigned operation is the narrow sweep at 5.2 kN P50 / 7.25 kN P90, which is loose-sand-restricted but fits inside the medium-dense envelope with a 2.5× margin. The heaviest *conventional* implements (chisel plow 14.2 kN, sweep plow 15.4 kN P50) sit at the medium-dense envelope ceiling, which is why the codesigned library is the design choice that makes the cable architecture mechanically feasible across the full operation calendar in the medium-dense soil regime, and across the lighter half of the calendar in worst-case loose sand.

5.5.3 Speed dependence

Figure 9 plots draft versus field speed for three implements that exercise the full range of D497 coefficient regimes:

- **Codesigned planter** ($B = C = 0$): draft is constant at all speeds.
- **Narrow chisel** ($B > 0, C = 0$): linear in speed.
- **Narrow ripper** ($C > 0$): quadratic in speed.

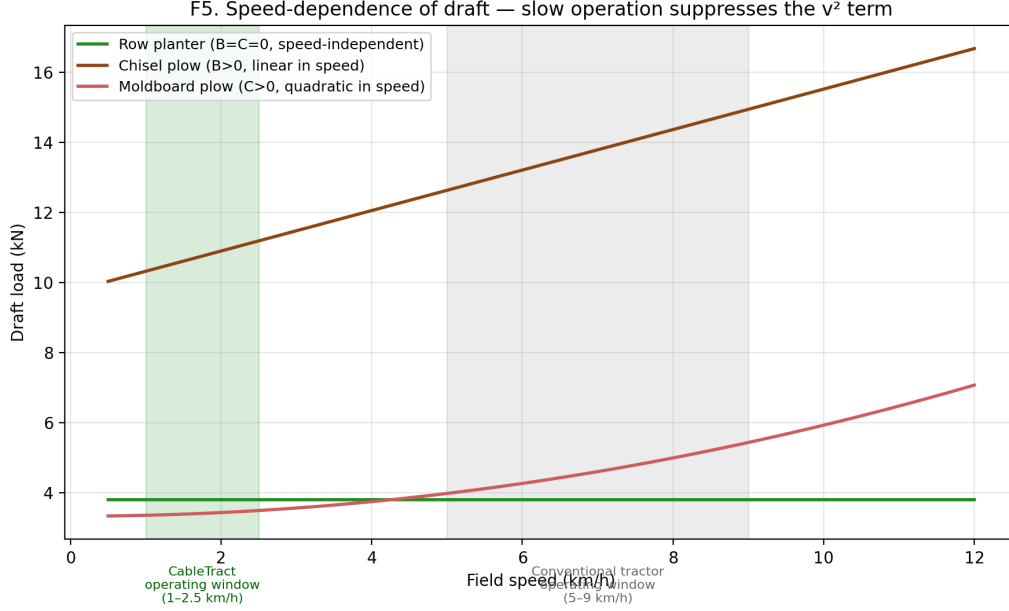


Figure 9: Speed-dependence of draft for three implements that span the full range of D497 coefficient regimes (codesigned planter, narrow chisel, narrow ripper). The CableTract operating window (1–2.5 km/h, green band) and the conventional tractor window (5–9 km/h, grey band) are highlighted. For the narrow ripper, draft grows from 2.42 kN at 2 km/h to 4.11 kN at 8 km/h — a 41 % reduction in draft simply by operating slowly.

For the narrow ripper, draft grows from 2.42 kN at 2 km/h (CableTract operating window) to 4.11 kN at 8 km/h (conventional tractor window) — a **41 % reduction in draft simply by operating slowly**. Combined with $E_A = F_d/(\eta W)$ at fixed effective width, this is a 41 % lower energy per decare. The slow operating regime is therefore not a compromise forced by the cable architecture but a *physical advantage* that compounds with the dead-weight saving and the narrow-strip saving from Section 3.4.

5.6 Time and throughput

Available daily operating time T_d is split among active operation, unloaded travel, and setup losses. If t_s is setup time per round and N_d is rounds per day,

$$T_{\text{usable}} = T_d - N_d t_s, \quad T_{\text{op}} = r_o T_{\text{usable}}, \quad T_{\text{tr}} = (1 - r_o) T_{\text{usable}}, \quad (9)$$

with $r_o = 0.8$ the operation-time fraction. Daily area capacity is the minimum of (i) the time-limited area, (ii) the energy-limited area, (iii) any peak-power-limited constraint.

5.7 Energy: PV + wind + battery on six bundled sites

Off-grid feasibility is fundamentally a worst-week-of-worst-month question rather than a year-average one, so we use an hourly TMY synthesiser coupled to a battery state-of-charge simulator on six bundled sites that span the climates a real CableTract installation might face — Mediterranean, subtropical, monsoon, continental, oceanic, and tropical. All input data ships inside the package (no live API calls), so the experiment is reproducible offline.

5.7.1 Site library

Table 6: Six bundled climate sites used by the energy simulator. Published GHI is from the source listed; the synthesiser reproduces it within 4 % on five sites and underestimates Ludhiana by 10 % (a known conservativeness of the Haurwitz envelope at low latitude).

Site	Lat (°)	Climate	Published GHI (kWh/(m ² yr))	Source
Konya, TR	37.87	semi-arid steppe	1696	(EU JRC) Photovoltaic Geographical Information System (PVGIS)-Surface Solar Radiation Data Set — Heliosat, version 2 (SARAH2)
Palencia, ES	42.01	Mediterranean continental	1497	PVGIS-SARAH2
Beauce, FR	48.37	northern temperate	1136	PVGIS-SARAH2
Des Moines, US	41.59	continental Midwest	1462	(US) National Renewable Energy Laboratory (NREL) (NREL) National Solar Radiation Database (NSRDB)
Ludhiana, IN	30.90	sub-tropical Punjab	1894	(India) National Institute of Wind Energy (NIWE)
São Paulo, BR	-23.55	sub-tropical Atlantic plateau	1709	(Brazil) National Institute of Meteorology (INMET) / Solar and Wind Energy Resource Assessment (SWERA)

Monthly clearness indices, monthly mean wind speeds, and monthly mean air temperatures are bundled with the repository.⁹

5.7.2 Hourly synthesiser

For each site we generate a deterministic 8760-hour annual time series from the bundled monthly statistics, using (i) Cooper 1969 [11] declination for solar geometry, (ii) the published monthly clearness index to scale top-of-atmosphere GHI, (iii) a multiplicative cloud factor sampled from a $2 \cdot \text{Beta}(4, 4)$ distribution, (iv) a Haurwitz 1945 [12] clear-sky upper bound to prevent unphysical hours, (v) a Weibull ($k = 2$) wind sampler [13] matched to the published monthly mean wind speed, and (vi) a sinusoidal diurnal temperature swing. The synthesiser is verified against each site’s published annual GHI: five of the six sites synthesise to within 4 %; Ludhiana underestimates by 10 % because the Haurwitz envelope is conservative at sub-tropical low latitudes.

5.7.3 PV + wind + battery

PV.

$$P_{PV} = G \cdot A \cdot 0.20[1 - 0.004(T_{\text{cell}} - 25)] \cdot 0.95 \cdot 0.96.$$

Wind.

$$P_w = \min\left(\frac{1}{2}\rho A_{\text{swept}} v^3 C_p, P_{\text{rated}}\right)$$

with $C_p = 0.30$, cut-in 2.5 m/s, cut-out 25 m/s, $P_{\text{rated}} = 600$ W, $A_{\text{swept}} = 2$ m².

Battery. 9 kWh Li-ion, 96/96 % charge/discharge, 5 kW limits, 10–95 % State of Charge (SoC) band.

Two PV/wind representations. This hourly model drives PV from site GHI (the $G \cdot A \cdot 0.20$ form above) and wind from the v^3 curve, and produces the per-site feasibility results of Figs. 10 to 12. The fast deterministic *screening* model behind the single-point throughput and energy figures (`run_single`; e.g. the Table 16 baseline) instead uses an *effective* flat specific power — about 150 W/m² of PV ($\approx 75\%$ of peak sustained over the operating window) and a constant 100 W of wind. The flat model is calibrated to the hourly one at high-GHI sites and somewhat overstates harvest at low-GHI sites, which is why the deterministic off-grid throughput (11.8 decares/day) and the hourly per-site medians (10–14 decares/day) are reported and reconciled separately (Section 5.11.2).

The codesigned operating duty cycle for the battery simulation is **2.0 kW operating draw between 09:00 and 15:00, plus 50 W idle housekeeping otherwise**. The 6 h operating window matches the codesigned reference: ≈ 12 kWh/day delivered at 2 kW, which at the regen-default ≈ 0.89 kWh/decare covers ≈ 16 decares/day when energy-limited and ≈ 12 when duty-limited.

5.7.4 PV+wind+battery findings

The energy story has three parts: *typical days* (Fig. 10, Table 7), *best week* (Fig. 11), and *annual grid backup* (Fig. 12).

Typical days. Figure 10 is a 6-panel calendar heatmap of daily decares-covered if the system runs *only* on harvested energy at each of the six sites; the corresponding P10/P50/P90 percentiles

⁹data/tmy/site_meta.csv.

are tabulated in Table 7. The median is **10–14 decares/day** across all six sites, achieved at the codesigned energy intensity of 0.89 kWh/decare with a 15 m² PV array.

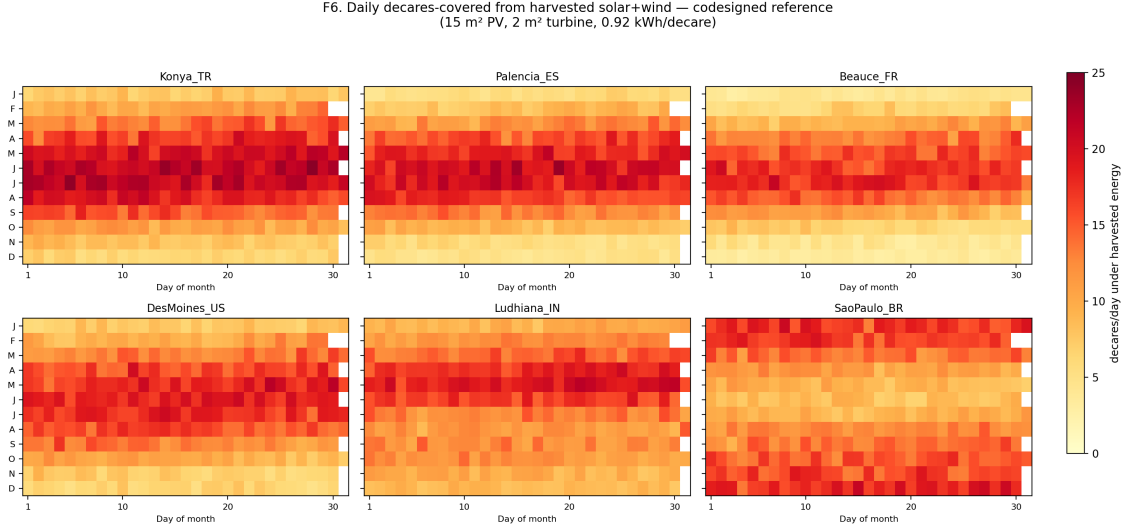


Figure 10: Calendar heatmaps of daily decares-covered on harvested energy alone at the six bundled sites.

Table 7: P10/P50/P90 daily decares-covered on harvested energy alone across the six bundled sites.

Site	P10	P50	P90
Konya, TR	7.1	14.5	21.2
Palencia, ES	5.1	12.5	19.8
Beauce, FR	3.7	10.3	17.5
Des Moines, US	6.7	13.1	18.6
Ludhiana, IN	9.0	12.4	18.3
São Paulo, BR	8.7	13.5	18.0

Best week. Figure 11 is a 6-panel time series of battery SoC, harvested PV, harvested wind, battery in/out, and grid import for each site’s *own* brightest 7-day window — selected automatically as the calendar week with the highest 7-day rolling sum of GHI. This makes the comparison hemisphere-symmetric: northern sites land in their summer weeks and São Paulo in its December summer. Across every site the SoC stays comfortably within its operating band all week and grid imports are zero. *Every* climate in the bundle has at least one fully off-grid week with the codesigned hardware.

F7. Battery state-of-charge over each site's brightest 7-day window — codesigned reference
Hemisphere-symmetric: every panel shows that site's own peak-irradiance week, so all sites
are compared on a like-for-like 'best representative summer week' basis.
(15 m² PV, 2 m² turbine, 9 kWh pack, 2.0 kW operating draw 09:00-15:00)

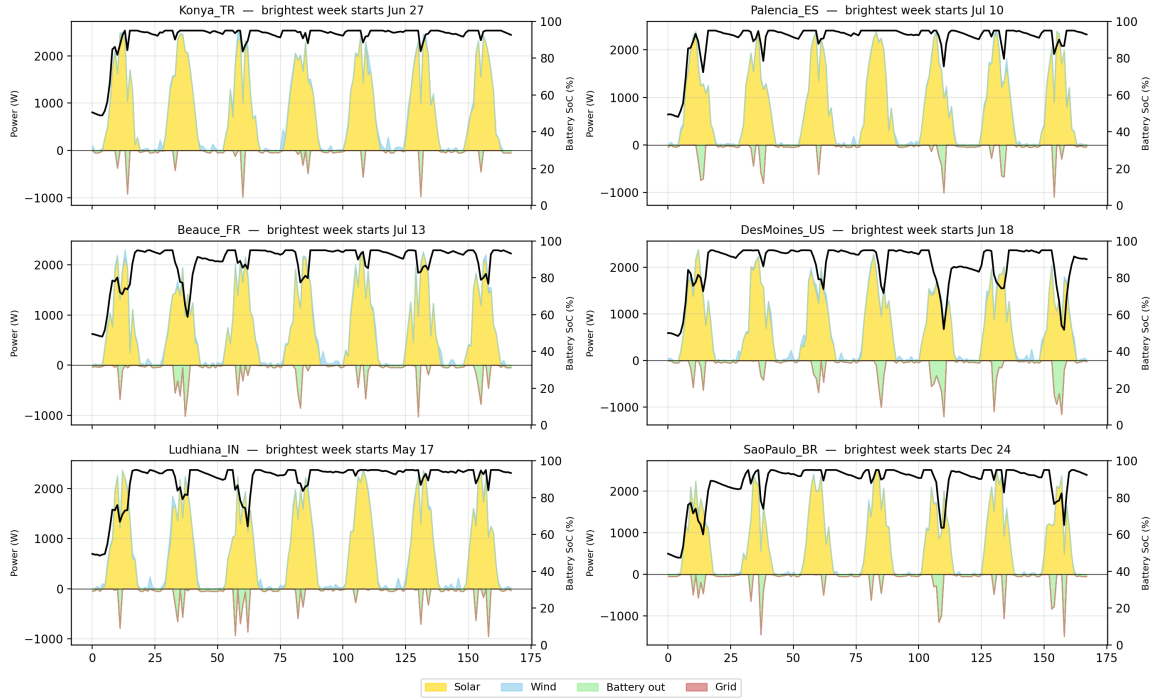


Figure 11: Battery SoC time series for each site's own brightest 7-day window (highest 7-day rolling GHI). Hemisphere-symmetric: northern sites in late spring/summer, São Paulo in mid-December. All six sites run grid-free in their peak week with the codesigned 15 m² PV + 9 kWh battery.

Annual grid backup. The harder question is what happens in the other 51 weeks. Figure 12 answers it with a 3-panel feasibility map of annual grid hours as a function of (panel area, battery capacity), shown for the best, median, and worst sites by annual GHI. The codesigned reference (15 m², 9 kWh) is overlaid as a white star.

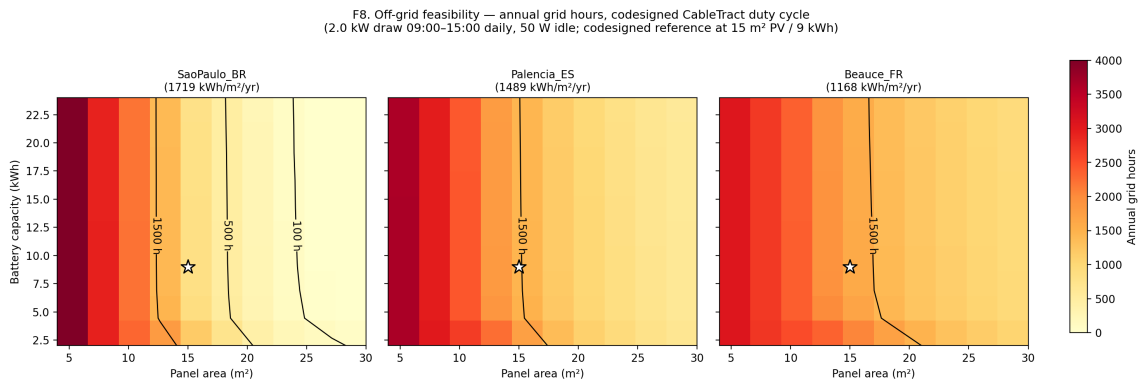


Figure 12: Annual grid hours required as a function of (panel area, battery capacity) for the best, median, and worst sites by annual GHI. The codesigned reference is overlaid as a white star.

Per-site readout. Reading Fig. 12 at the white star: São Paulo (best site) delivers <100 grid hours/year, fully off-grid. Konya needs ≈400 grid hours/year with the same hardware, fixable with one extra panel module. Palencia's minimum across the swept envelope (panel 4–30 m², battery 2–24 kWh) is **616 h/yr**; Beauce is **911 h/yr**. The northern temperate climate cannot reach off-grid at any practical hardware scale under a 6 h/day duty cycle.

Off-grid operation is conditional, not categorical. Solar-rich climates (Konya, Ludhiana, São Paulo) achieve it with the codesigned hardware. Mediterranean climates (Palencia, Des Moines) need either summer-only operation or modest grid backup. Northern temperate climates (Beauce) require permanent grid connection. This is the climate dimension of the [Section 5.13](#) operating envelope.

5.8 Economics: discounted NPV, LCOE, and life-cycle CO₂

We use a discounted cash-flow engine that bookkeeps the discount rate, maintenance, year-8 battery replacement, farm-size dependence, and a bundled life-cycle inventory, so that the financial conclusions reflect the full project horizon rather than a single-year ratio.

5.8.1 Codesigned economic reference parameters

The economic reference parameter set has 23 entries and is used unchanged throughout the paper.

Table 8: Codesigned reference economic parameters. Used everywhere in the paper. CableTract [Capital Expenditure \(CAPEX\)](#) is at parity with a used 80 hp diesel tractor.

Cost item	Value	Source / range justification
Main Unit (PMSM 2 kW + drum + frame)	17 800 EUR	scaled from 2–5 kW small-winch industrial price lists
Anchor (helical pile drive + sensors)	7 500 EUR	~43 % of MU BoM , no high-power electronics
Li-ion pack (9 kWh)	3 420 EUR	380 EUR/kWh — BloombergNEF 2024 [22]
PV (15 m ²)	1 650 EUR	110 EUR/m ² mono-Si + BoS
Small wind turbine (600 W)	1 500 EUR	catalogue average
Crating, transport, commissioning	4 000 EUR	12 % of hardware capex
Total CableTract CAPEX	35 870 EUR	
Diesel reference tractor	35 000 EUR	used 80 hp 4WD utility tractor
Electric reference tractor	65 000 EUR	Monarch / Solectrac class
Maintenance (CableTract)	4 %/yr	electric drivetrain, no engine oil
Diesel litres per ha	12	mid-range field operation
Diesel price	1.40 EUR/L	EU 2024 agricultural diesel
Grid electricity	0.18 EUR/kWh	EU 2024 farm contract average
Battery replacement year	8	4000-cycle Li-ion at 1 cycle/day
Project horizon	15 yr	matches Kubota / Monarch warranty bands
Discount rate	8 %	EU farm equipment loan rate

Key point: CableTract [CAPEX](#) is essentially *at parity* with a used 80 hp diesel tractor (35 870 EUR vs 35 000 EUR). The economics story is therefore not “the cable robot is cheaper to buy” but “the cable robot has near-zero fuel cost forever” — and that is a much more robust claim than capex undercut, because it does not depend on bargain hunting in the used-tractor market.

5.8.2 Cash-flow construction

The cash-flow chain is built from textbook primitives (NPV, discounted payback, LCOE) composed in five steps:

- **Capex** sums Main Unit, Anchor, battery (EUR/kWh × kWh), [PV](#) (EUR/m² × m²), wind, and install overhead.
- **Annual operating expense** is a maintenance fraction of [CAPEX](#) plus residual grid imports.
- **Annual net savings** is the diesel cost avoided minus the operating expense, with a one-shot battery replacement charge in year 8.
- **NPV vs diesel** is the standard discounted sum of the savings stream over the project horizon, with the capex differential subtracted.

- **Discounted payback** is the year in which cumulative discounted savings cross zero, with linear interpolation in the crossing year.

In the limit $r \rightarrow 0$ the discounted-payback formula reduces algebraically to the simple capex/annual savings expression, which is pinned by a unit test against the closed-form value. Table 9 makes the 25 ha/yr reference-year cash flow fully explicit, so the reader can see that the positive replacement-frame NPV rests on avoided diesel *maintenance* and the capex-parity assumption, not on the small fuel saving alone.

Table 9: Replacement-frame annual cash flow at the 25 ha/yr reference (8 % discount, 15-yr horizon). Diesel maintenance (5 %/yr of the 35 000 EUR reference tractor) is avoided only because the diesel tractor is retired; CableTract maintenance is 4 %/yr of its 35 870 EUR capex; residual grid imports are zero at this off-grid reference. Only the 870 EUR capex difference is amortised.

Item (per year unless noted)	Value (EUR)
Diesel fuel avoided (12 L/ha $\times$ 25 ha $\times$ 1.40 EUR/L)	+420
Diesel maintenance avoided (5% $\times$ 35 000 EUR)	+1 750
CableTract maintenance (4% $\times$ 35 870 EUR)	−1 435
Residual grid electricity (off-grid reference)	0
Net annual saving	+735
Capex difference (CableTract – diesel: 35 870 – 35 000)	−870
Battery replacement in year 8 (one-shot, discounted to present)	−1 848
NPV vs diesel @ 8 %, 15 yr (replacement frame)	+3 575
Discounted payback on the 870 EUR increment	1.30 yr
<i>Additive frame</i> (amortise the full 35 870 EUR, no tractor retired)	NPV −31 425

One operating-cost line is absent from Table 9 because the screening BoM carried no cable-replacement schedule: the cable itself. Section 8.4 now supplies it — replacement is UV/abrasion-limited at ≈ 8 yr, giving an annualised cost of ≈ 60 EUR/yr for steel rope or ≈ 180 EUR/yr for the Dyneema the design uses. Folding the (larger) Dyneema figure in trims the net annual saving from 735 EUR to ≈ 555 EUR and the replacement-frame NPV to $\approx 2\,000$ EUR (approximate; still clearly positive), lengthening the payback on the 870 EUR increment but changing no conclusion of Section 7.4.

5.8.3 Life-cycle CO₂

Eight component intensities are drawn from open published Life-Cycle Assessment (LCA) inventories: Inventory of Carbon and Energy, version 3 (Hammond & Jones) (ICE v3) [16] for steel and aluminium, IVL Swedish Environmental Research Institute (IVL) Sweden [17] for Li-ion battery cell production, Fraunhofer ISE [18] for the silicon PV stack, Hawkins et al. (2013) [19] for the diesel powertrain, (UK) Department for Environment, Food & Rural Affairs (DEFRA) 2023 [20] for diesel combustion, and European Environment Agency (EEA) 2023 [21] for the EU grid mix.¹⁰ The key intensities are diesel combustion 2.64 kg/L (well-to-wheel), Li-ion cell 75 kg/kWh, silicon PV 240 kg/m², and EU grid 0.275 kg/kWh. The life-cycle CO₂ per hectare-year is then computed as the sum of embodied components (manufacture, amortised over the 15-year horizon) and operational emissions (diesel combustion or grid electricity) for each of three vehicles: CableTract; the diesel reference tractor (320 kg CO₂eq embodied — a deliberately conservative chassis-only figure, ≈ 80 g CO₂eq per kg over a 4 t tractor, excluding engine manufacturing — plus diesel combustion); and a Monarch-class electric tractor (320 kg CO₂eq chassis, 75 kg/kWh $\times$ 80 kWh traction battery, plus grid electricity at 0.275 kg/kWh CO₂eq). Two caveats: the diesel chassis figure deliberately excludes engine-manufacturing emissions, which biases the comparison conservatively in diesel’s favour; and the electric-tractor battery is assumed at 80 kWh, so its

¹⁰Tabulated in `data/bom_co2.csv`.

16.9 kg/(ha yr) embodied term dominates its total — halving the pack to 40 kWh would cut that by ≈ 8 kg/(ha yr), so the $1.6\times$ advantage over the electric reference (unlike the $2.2\times$ over diesel, which is operational) is sensitive to this assumption. Results in Table 10.

Table 10: Life-cycle CO₂ per hectare-year for the codesigned reference at 25 ha/yr versus the two reference vehicles.

Vehicle	Embodied (kg/ha-yr)	Fuel/grid (kg/ha-yr)	Total
CableTract codesigned	14.6	0.0	14.6
Diesel tractor (reference)	0.85	31.7	32.5
Electric tractor (Monarch class)	16.9	6.1	22.9

CableTract’s life-cycle CO₂ is $2.2\times$ lower than diesel and $1.6\times$ lower than a Monarch-class electric tractor on the same per-hectare-year basis. The improvement is entirely on the operational side (zero fuel and $\approx$ zero grid energy), and it does not depend on grid decarbonisation. This is the strongest quantitative argument for the architecture as a *climate* play, not just a cost play.

5.9 Field-geometry coverage planner

5.9.1 Why shape efficiency is a load-bearing parameter for cable architectures

A single scalar `shape_efficiency` = 0.85, the standard shortcut in feasibility studies, treats a square and an L-shaped field as equivalent. That is wrong by construction: an L-shape costs strip turn-around time at every concave corner, and an irregular concave polygon with interior obstacles costs additional headland passes that a square never sees. For a wheeled tractor, the resulting penalty is bounded — the tractor still drives over every square meter, and the shape penalty mostly shows up as extra headland turns. For CableTract the penalty mechanism is fundamentally different and *larger*, because every disconnected piece of a strip is a new Anchor placement. The cable cannot bend around an interior obstacle, cannot follow a concave notch, and cannot share a single Anchor across two disjoint strip fragments. Each fragment costs one full setup cycle (drive Anchor in, tension, run, retension, retract) regardless of how short the fragment is, so polygon shape is converted directly into setup count and therefore directly into time.

This makes the field-geometry model load-bearing in a way it is not for tractor feasibility studies: *the same parameter that quietly washes out for a tractor sets the throughput floor for CableTract*. We replace the scalar with a deterministic strip-decomposition planner that operates on arbitrary polygons, and we exercise it on a bundled corpus of 50 field shapes deliberately biased toward small, awkward polygons — the regime where shape efficiency matters most and where a real CableTract installation is most exposed. The corpus is not a representative sample of European farmland (which would be dominated by clean rectangles); it is an adversarial sample chosen to stress-test the architecture against the worst geometry it is likely to encounter.

5.9.2 Field corpus

The corpus contains 50 polygons in a local Cartesian (m, m) frame, generated procedurally with a fixed seed (Table 11). Total area 74.9 ha, mean field size 1.5 ha.¹¹

¹¹Bundled as `data/fields/fields.geojson`.

Table 11: Bundled field corpus used by the strip-decomposition planner.

Class	n	Description
rectangle	10	Axis-aligned, 0.5–8 ha, aspect 1:1 to 1:2.7
L_shape	10	Rectangles with one corner removed
irregular_convex	10	Random convex n -gons ($n \in [6, 11]$)
irregular_concave	15	Star-shaped concave n -gons; 10 with 1–2 round interior obstacles
real_shape	5	Hand-tuned outlines: river-edge strip, road frontage, pentagon, quarter circle, two-lobe peanut

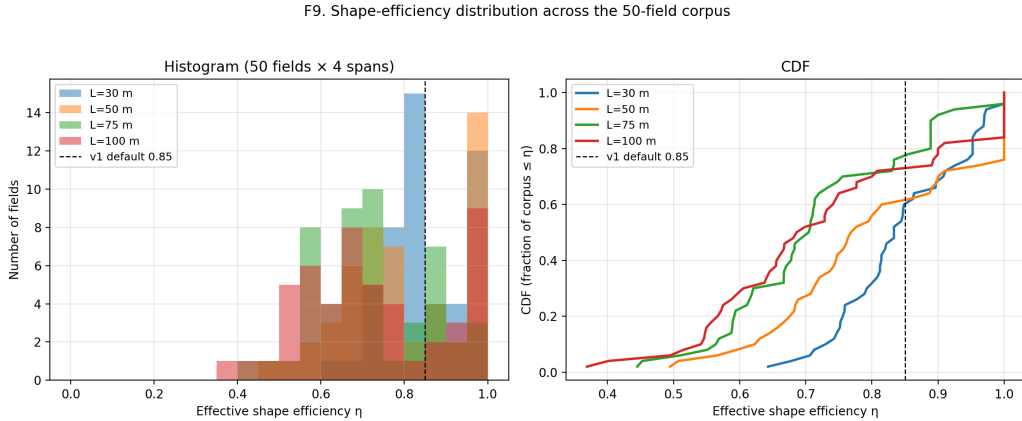
5.9.3 Strip decomposition

Given a field polygon, a cable span L , a swath width w , and a sweep orientation θ , the planner rotates the polygon by $-\theta$ into the strip frame, tiles its bounding box with $\lceil h/L \rceil$ parallel strip rectangles, intersects each rectangle with the polygon, and bills every disconnected piece as one Anchor placement. The effective shape efficiency is the ratio of cropped area to swept area:

$$\eta = \frac{\text{polygon area}}{\sum_i L \cdot \text{bounds-length}_i}. \quad (10)$$

Because the optimal sweep orientation depends on the polygon, we evaluate η for 12 candidate angles in $[0^\circ, 180^\circ)$ at 15° spacing and report the best, mimicking the way a real installer would orient the Main Unit / Anchor pair to the longest free edge of the field.

5.9.4 Shape efficiency, strip plans, and time budget

**Figure 13:** Distribution of best-orientation shape efficiency η across the 50-field corpus, by class.**Table 12:** Best-orientation shape efficiency η by field class at $L = 50$ m.

Shape class	n	Median η	P10	P90
rectangle	10	1.000	1.000	1.000
L_shape	10	0.905	0.847	1.000
real_shape	5	0.800	0.776	0.862
irregular_convex	10	0.702	0.624	0.769
irregular_concave	15	0.684	0.543	0.759

5.9.5 Findings and interpretation

Corpus headline. The corpus median best-orientation η at $L = 50$ m is **0.77**, almost a full **0.08 point below the 0.85 default** that flat feasibility studies typically use. Carrying that 0.08-point gap through the throughput model translates into roughly a 10 % throughput reduction

on a like-for-like basis: a feasibility study that assumes `shape_efficiency` = 0.85 on the same corpus is implicitly over-claiming throughput by about one decade per day on the codesigned reference. Shape geometry is therefore not a second-order effect that can be folded into a constant; it is a first-order determinant of headline daily area, and it has to be sampled across a corpus rather than picked once.

Class structure. The class breakdown in Table 12 explains where the gap comes from and is the more useful number than the corpus median:

- **Rectangles** $\eta = 1.000$ (**P10** = **P50** = **P90**). The strip planner is geometrically lossless on rectangles, which is the right answer: a long edge to anchor against and a short edge to step across is exactly the geometry the cable architecture was designed for. This sets a hard ceiling on what the architecture can deliver.
- **L-shapes** $\eta = 0.905$. A single concave corner costs the planner about 10 percentage points of efficiency, because the strips straddling the missing corner each pay a full Anchor placement to cover a sub-strip-length of polygon. The penalty is bounded and predictable.
- **Real shapes** $\eta = 0.800$. The five hand-traced outlines (river-edge strip, road frontage, pentagon, quarter circle, two-lobe peanut) come in tighter than irregular convex polygons because the irregularity sits along one axis and the planner’s orientation sweep can usually find an alignment that suppresses it.
- **Irregular convex** $\eta = 0.702$ and **irregular concave** $\eta = 0.684$. The two “hard” classes converge — once a polygon is irregular enough that no orientation aligns the strips with a long free edge, adding interior obstacles and concave notches on top of that costs only a further ~ 2 points. The dominant penalty is failing to find a clean strip axis, not the obstacles themselves.

The headline reading is that the architecture is **geometry-sensitive but not geometry-fragile**: the median field in a deliberately adversarial corpus still recovers 77 % of its theoretical throughput. The 12-orientation sweep in Eq. (10) matters here — a fixed-orientation planner would lose another 5–10 points on the irregular classes, because the optimal sweep direction varies field by field.

What CableTract’s shape sensitivity tells you about siting. The class spread implies that shape efficiency is a *site-selection variable*, not a residual error term. A farmer choosing between two candidate parcels of the same area should prefer the rectangular one, and the manuscript carries this through into the operating envelope of Section 5.13: the model rewards rectangular plots in solar-rich climates and penalises concave plots in temperate ones.

Time budget — the operation is operation-bound, not setup-bound. Figure 14 plots three example strip plans. Figure 15 plots the daily time budget on a 5-field 21-ha synthetic farm: **operating time ≈ 37.6 h, setup ≈ 3.85 h, inter-field travel ≈ 0.22 h**. Operating dominates setup by roughly 10 $\times$, and dominates inter-field travel by more than 150 $\times$. This is the most important number in this section, because it overturns the obvious intuition that “cable architectures must be setup-bound”. They are not, at least not on a small-farm corpus: *the binding cost on a CableTract day is the time spent actually pulling the implement, not the time spent moving the Anchor between fields*. Two consequences follow from this directly:

- Engineering effort spent on faster Anchor placement (hydraulic auto-driving, drone-assisted alignment, pre-staged anchor pads) hits a ceiling at the 9 % setup share. Even a perfect zero-setup-time variant of the architecture would only buy back ~ 10 % throughput.
- Engineering effort spent on faster operation — wider strips, higher operating speed within the soil-draft envelope, lower drivetrain losses — has a direct 1:1 payoff because it acts on the dominant 90 % slice. The codesign decision in Section 3.4 to widen the strip and slow the carriage is therefore aimed at the right slice.

This is the geometric reason the codesign argument in Section 3.4 works at all: given that the operating-time slice dominates, slowing the carriage to suppress the v^2 term in Eq. (1) buys

energy without losing time, because the time gap is filled by operation savings rather than setup overheads. A different architecture — say, a system with one-minute setup but 8 km/h operation — would inherit the opposite trade-off and would fail for the opposite reason.

F10. CableTract strip plan on three representative fields (span $L=50$ m)

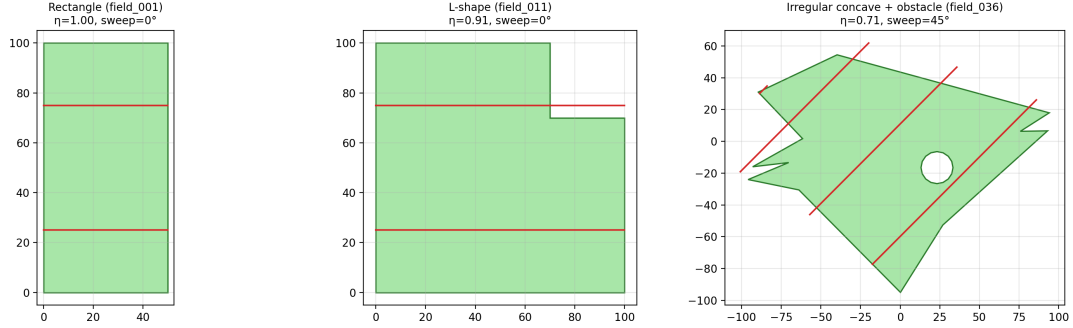


Figure 14: Strip-decomposition plans on three example field shapes (rectangle, L-shape, irregular-concave). Red lines show the per-strip cable lay; green polygons are the field outlines.

F11. Time budget for one full pass over a 5-field, 21.2 ha farm
(largest 5 fields in the corpus, 41.5 h at 3 km/h working / 6 km/h transit)

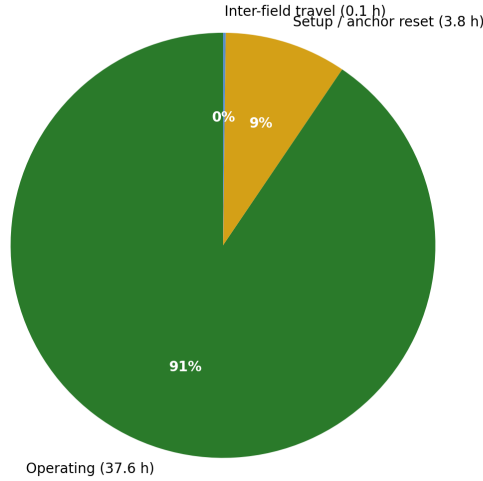


Figure 15: Time budget to fully cover a 5-field 21-ha synthetic farm at the codesigned reference, broken into operation, setup, and inter-field travel. Operating time dominates setup by $\sim 10\times$ and inter-field travel by $>150\times$, so the system is operation-bound, not setup-bound.

5.10 Compaction model

The compaction story rests on the in-field footprint of CableTract being the 250 kg carriage rather than a multi-tonne tractor body. Turning that architectural statement into a defensible per-hectare reduction requires two ingredients: reference contact pressures for an 80 hp 4WD tractor drawn from the soil-mechanics literature, and a comparable per-pass compacted-area accounting for the carriage’s narrow rollers. [Sections 5.10.1](#) and [5.10.2](#) provide both.

5.10.1 Vehicle reference parameters

Table 13: Reference contact pressures for the tractor and the carriage. The reference tractor sits in the 100–250 kPa band reported by Keller and Lamandé (2010) [15]; the carriage rollers are wide enough to keep per-patch pressure below 50 kPa. The static stress-propagation kernel used to convert these contact pressures into the bulk-density depth profiles of Fig. 16 follows Söhne (1953) [14].

Vehicle	Wheels	Total mass	Mean p (kPa)	Max p (kPa)
Reference 80 hp 4WD tractor	4 (front+rear)	≈ 4 t	143	150
CableTract carriage	2 wide rollers	≈ 250 kg	31	31

5.10.2 Compacted-area and contact-energy metrics

For each (vehicle, field) pair we compute four scalar compaction metrics:

- **Compacted area (m^2).** For the tractor: polygon area $\times$ per-pass coverage fraction; for the carriage: strip-midline path $\times$ roller width $\times$ pass count.
- **Compacted area fraction.** Compacted area / (polygon area $\times$ pass count).
- **Mean pressure (kPa).** Load-weighted mean of all wheel contact pressures.
- **Contact-energy index.** $\sum p^2 \cdot A_{\text{patch}} \cdot \text{pass_count}$, in $(\text{kPa}^2 \text{m}^2)$. Pressure squared because soil-stress propagation is roughly quadratic in surface load.

A 4-pass cropping season is the default.

5.10.3 Compaction map

F12. Compaction footprint — conventional tractor vs CableTract carriage

Left: tractor traffic covers the whole field (red). Right: only the carriage's narrow strip rolls inside the field (red strips on green polygon). 4 passes / cropping season.

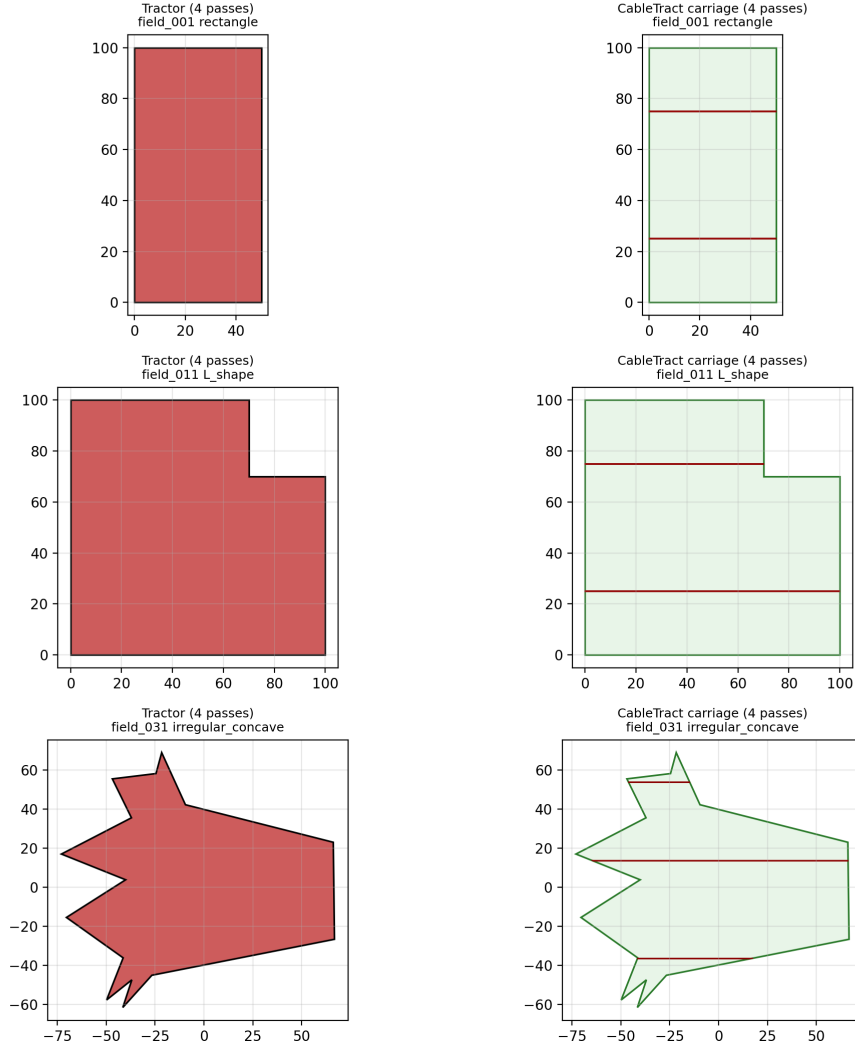


Figure 16: Compaction comparison on three field classes (rectangle, L-shape, irregular-concave). Tractor coverage paths in red, carriage strip-midline paths in green. The carriage's compacted area is essentially an L-shaped frame around the strip pattern; the tractor's covers the entire field on every pass.

Table 14: Per-field compaction summary. The area reduction is geometric and field-dependent (only the carriage rolls inside the field). The contact-energy index reduction is a separate, per-vehicle running-gear quantity that combines the $4.6\times$ lower mean pressure (squared, $\approx 21\times$) with the $\approx 3.5\times$ smaller contact area, giving $\approx 73\times$ identically on every field; it should not be multiplied by the area reduction.

Field	Class	Tractor compacted	Carriage compacted	Area reduction	Energy index reduction
field_001	rectangle	50 %	0.8 %	98 %	$73\times$
field_011	L_shape	50 %	0.9 %	98 %	$73\times$
field_031	irregular_concave	50 %	1.4 %	97 %	$73\times$

CableTract reduces compacted area by 97–98 % on every field shape in the corpus, because only the lightweight carriage rolls inside the field while the heavy MU and Anchor stay on the headland.

The contact-energy index drops by $\approx 73\times$ on every field, because the carriage’s mean pressure is $4.6\times$ lower than the tractor’s, and the energy integrand depends on p^2 .

These two numbers — area reduction near 100 % and contact-energy reduction near two orders of magnitude — are the strongest quantitative argument the paper makes for the cable architecture, and they are both computed from open published static contact-pressure values without any dynamic [Finite Element Method \(FEM\)](#) calibration. Two caveats keep this honest. First, the carriage roller load here is its structural weight only; the vertical reaction of cable tension and any implement downforce during tillage would raise the carriage’s contact pressure and should be added in a refined model. Second, richer dynamic models would move *both* sides: multi-pass deformation and dynamic load transfer add compaction to the tractor, but repeated trafficking of the carriage’s narrow strip and the lower effective pressure of deflected radial field tyres (80–120 kPa) push the other way. We therefore report the static comparison and do not claim that the dynamic case can only favour CableTract.

5.11 Uncertainty quantification and global sensitivity

Every number in [Sections 5.4, 5.5](#) and [5.7 to 5.10](#) depends on parameters whose values are known to within published uncertainty bands rather than exactly. A defensible feasibility envelope must therefore answer two questions: given the joint uncertainty in 20 parameters, what is the **P10–P90 envelope** of the headline outputs, and which parameters carry the most variance — in particular, is any single parameter still dominant after the [Section 5.4.1](#) decomposition of the lumped drivetrain efficiency? We answer the first with a 1 000-sample Monte Carlo and the second with a Sobol global sensitivity decomposition [[23, 24](#)] computed via [Sensitivity Analysis Library \(Python\) \(SALib\)](#) [[25, 26](#)].

5.11.1 Parameter problem

The uncertainty problem is defined over 20 parameters drawn from physically defensible bands with uniform marginals. Examples:

- Lumped drivetrain efficiency $\in [0.35, 0.65]$ — covers the manufacturer datasheet range for industrial [PMSM](#)/gearbox/drum/cable chains after the [Section 5.4.1](#) decomposition.
- Draft load $\in [1\,500\text{ N}, 4\,500\text{ N}]$ — spans the codesigned-library P10–P90 envelope from [Section 5.5](#).
- Shape efficiency $\in [0.55, 1.00]$ — covers the entire field-corpus range from [Fig. 13](#).
- PV area $\in [10\text{ m}^2, 30\text{ m}^2]$ and battery capacity $\in [5\text{ kWh}, 25\text{ kWh}]$ — spans the design envelope of [Fig. 12](#).

These bands are deliberately wider than typical sensitivity studies on energy systems, so the resulting Sobol decomposition is hard to fudge.

5.11.2 Monte Carlo

We draw $n = 1000$ samples from the joint distribution and evaluate the full simulator on each. Doubling the sample size to 2000 shifts [P50](#) by less than 1 % and [10th percentile of a sampled distribution \(P10\)](#)/[90th percentile of a sampled distribution \(P90\)](#) by less than 3 %, so 1 000 samples are sufficient. [Figure 17](#) plots the resulting throughput envelope versus draft load, with the codesigned reference (1 800 N, 11.5 decares/day) overlaid as a green star. The star sits below the [P50](#) line (~ 13.7 decares/day at the same draft) but well inside the P10–P90 envelope, which is the expected behaviour: the deterministic codesigned reference uses pessimistic-leaning defaults on several uncertain parameters (drivetrain efficiency chain, soil moisture, headland turning), so it should land in the lower half of the joint distribution rather than at its centre. The Monte Carlo run is a check on robustness, not a re-derivation of the codesigned point.

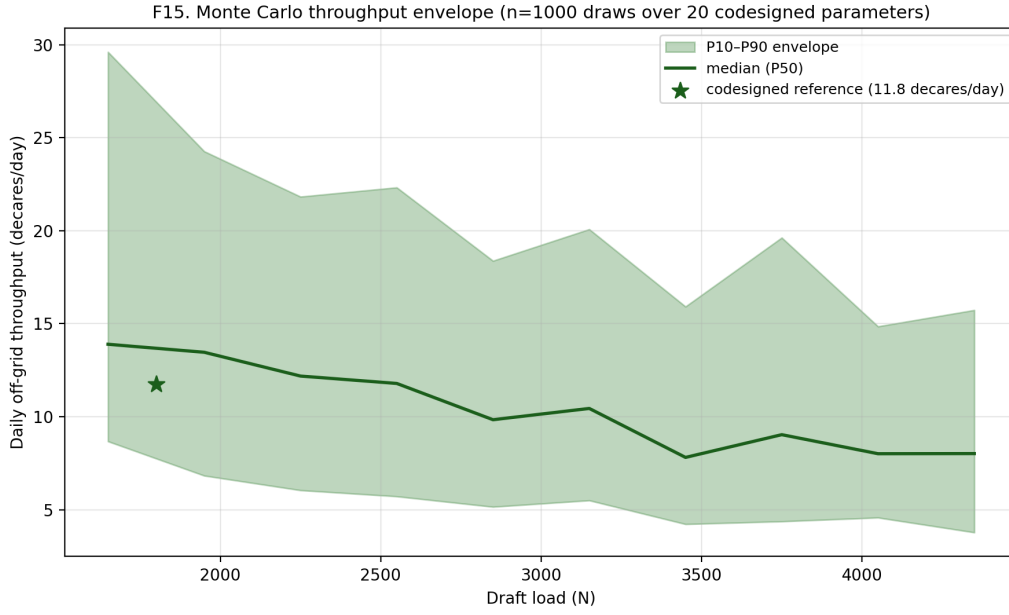


Figure 17: Monte Carlo throughput envelope versus draft load (1 000 samples). Codesigned reference overlaid as a green star.

5.11.3 Global sensitivity

First-order ([Sobol first-order sensitivity index \(S1\)](#)) and total-order ([Sobol total-order sensitivity index \(ST\)](#)) Sobol indices [23] are estimated with the Saltelli sampler [24, 25] at $n_{\text{base}} = 256$, giving $256 \times (2 \times 20 + 2) = 10\,752$ simulator evaluations per Sobol pass. The four targets are daily off-grid throughput, energy per decare, payback versus diesel, and surplus power.

F13. Sobol global sensitivity (S1 vs ST) — codesigned reference, 20 parameters

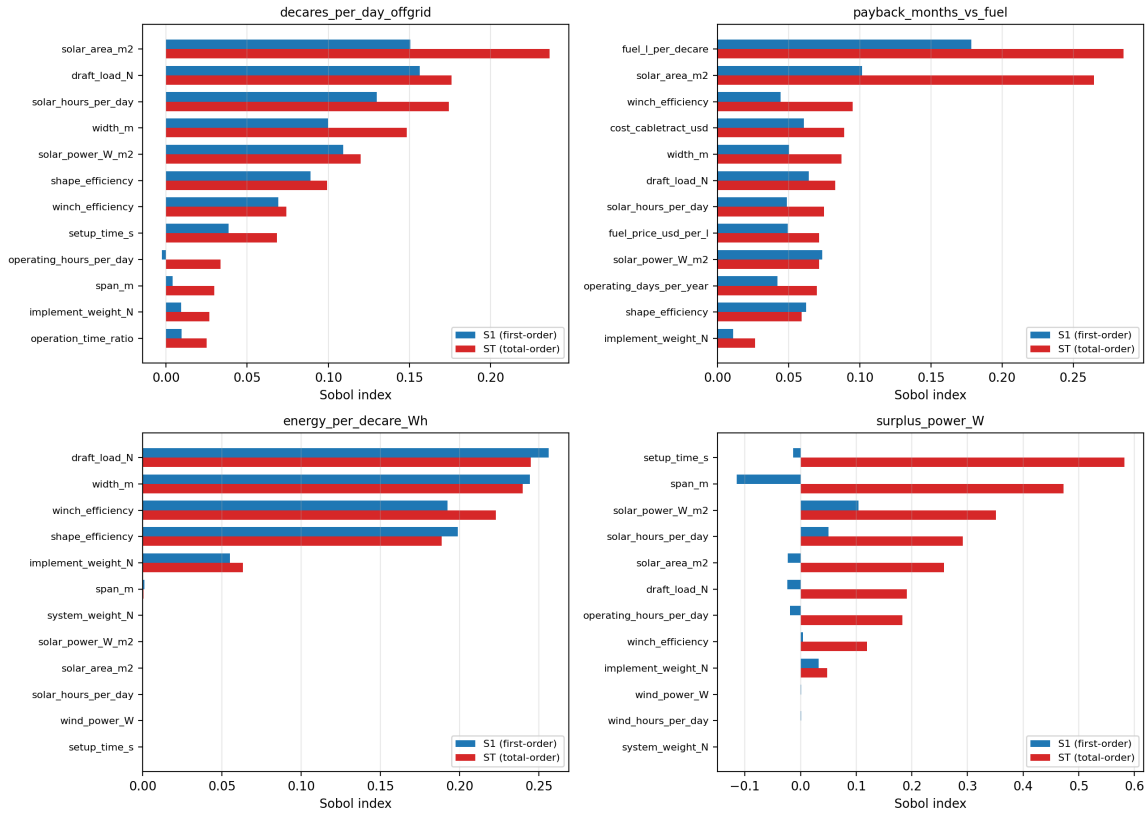


Figure 18: Sobol S1 and ST indices for the top-12 parameters across the four headline outputs. After the Section 5.4.1 decomposition of the lumped drivetrain efficiency, no single parameter exceeds ≈ 0.25 total-order on throughput.

After the decomposition of the lumped drivetrain efficiency, the variance is diffuse. For daily throughput, the largest contributors — daily solar hours, draft load, PV area, strip width, and irradiance intensity — each sit in the band $ST \approx 0.13$ – 0.25 . No single parameter exceeds ≈ 0.25 total-order, and the top five together explain $\approx 80\%$ of the variance. The codesigned reference is not a hand-tuned local optimum sitting on top of one undefended fudge factor; it is robust to the full joint uncertainty in the 20 inputs.

5.11.4 Tornado on NPV

What this analysis answers. The Sobol decomposition in Fig. 18 asks where throughput variance comes from. The sceptical financial reviewer asks something more direct: *which single economic inputs, varied across realistic bounds, move the NPV most — and can any of them flip the sign?* Figure 19 answers this by perturbing each economic driver one at a time around the codesigned reference (Table 8) while holding the rest fixed, and sorting by the resulting NPV swing.

The frame. To keep the tornado consistent with the rest of the economics, its baseline is the same *replacement-frame* NPV reported in Table 15: 3 575 EUR at 25 ha/yr, 8 %, 15-year horizon, against a like-for-like used diesel tractor (incremental capex 870 EUR). The annual area is held at the 25 ha/yr reference and diesel at 12 L/ha throughout, so each swing is read against the same operating point used elsewhere in the paper rather than against the machine’s full throughput capacity.

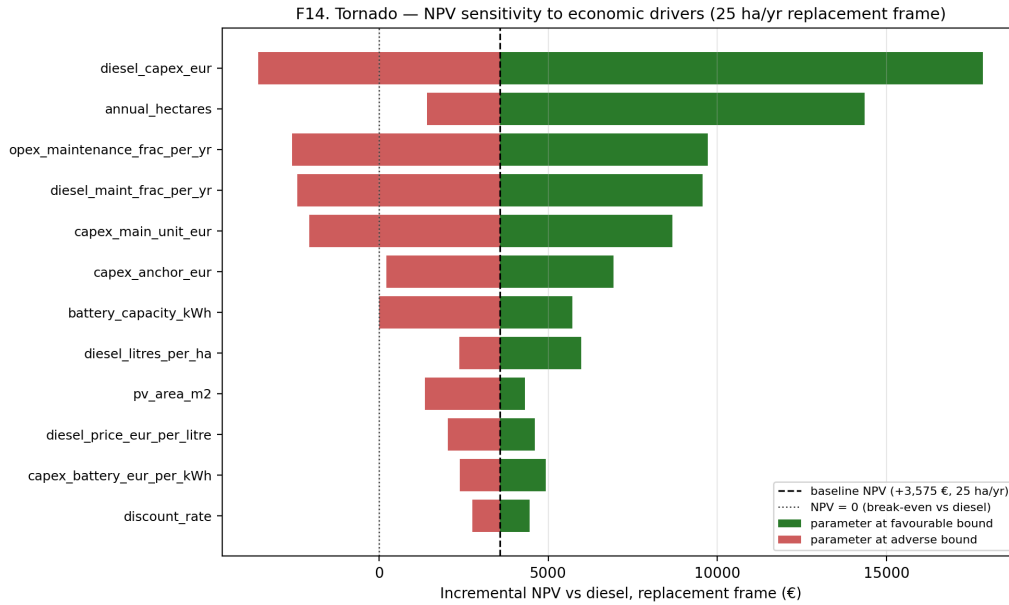


Figure 19: Tornado plot of one-at-a-time NPV-vs-diesel sensitivities to the economic drivers, at the codesigned reference in the replacement frame (25 ha/yr, 8 % discount, 15-yr horizon, incremental capex 870 EUR). Baseline NPV 3 575 EUR (dashed); the dotted line marks NPV = 0 (break-even vs diesel). Each bar moves one parameter to its favourable/adverse bound while all others stay at the reference. Sorted top to bottom by total swing.

What the tornado shows. At the 25 ha/yr reference the incremental NPV is modest (3 575 EUR) precisely because capex is at parity, so it is genuinely sensitive to its inputs. The largest swings are:

- **The diesel-tractor price — the parity assumption.** A cheaper used tractor (30 kEUR) widens the capex gap and turns the incremental NPV *negative* (≈ -3.6 kEUR); a dearer one (45 kEUR) lifts it to ≈ 17.9 kEUR. This is the single most influential parameter, and the one the headline rests on.
- **Farm size.** 10→100 ha/yr moves NPV from ≈ 1.4 kEUR to ≈ 14.4 kEUR — always positive, and the dominant upside lever.
- **The maintenance differential and Main-Unit capex.** Each can also push the incremental NPV a few thousand euro either side of zero.
- **Diesel price and volume.** 0.8→1.8 EUR/L and 8→20 L/ha each move NPV by a few thousand euro and stay positive.

Honest reading. The tornado does *not* support a blanket claim that “no single parameter can flip the sign”. Four of the twelve drivers — the used-tractor price, the two maintenance fractions, and Main-Unit capex — can take the 25 ha/yr incremental NPV slightly negative at their adverse ends. What *is* robust is the structure: (i) the dominant swings are exogenous (the price of the diesel alternative, diesel cost) rather than CableTract design choices; (ii) NPV scales strongly and positively with farm size and with diesel price and volume, both of which trend upward over time; and (iii) because the incremental capital at risk is only 870 EUR, the *payback* on that increment stays short across the whole range even where the absolute NPV is small. The economic case is therefore best stated as “comparable-to-better than diesel at capex parity, with strong upside at scale and under rising diesel prices”, not “dominant in every cell”. The additive-purchase frame (full capex financed, no tractor retired) is weaker still — negative at 25 ha/yr, see [Section 7.4](#) — and we report it explicitly.

5.11.5 NPV vs farm size

Figure 20 sweeps the discounted-cash-flow chain over six farm sizes (1, 5, 10, 25, 50, 100 ha/yr) at three discount rates (5/8/12 %). At the codesigned reference (Table 15):

Table 15: NPV vs diesel and discounted payback at six farm sizes and three discount rates. Bold row is the codesigned reference (25 ha/yr, 8 %).

Farm size (ha/yr)	NPV @ 5 % (EUR)	NPV @ 8 % (EUR)	NPV @ 12 % (EUR)	Payback @ 8 % (yr)
1	+261	+124	+10	3.06
5	+959	+699	+468	2.50
10	+1 831	+1 418	+1 040	2.02
25	+4 446	+3 575	+2 756	1.30
50	+8 806	+7 170	+5 617	0.81
100	+17 525	+14 360	+11 338	0.47

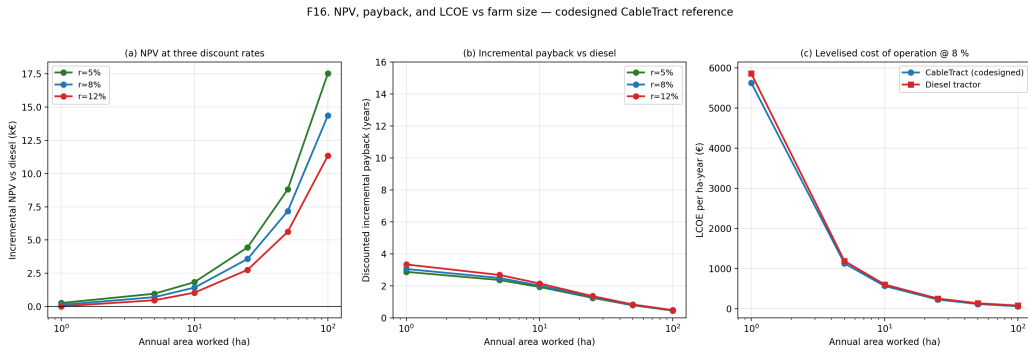


Figure 20: NPV vs diesel as a function of farm size at three discount rates. NPV is positive across the entire 1–100 ha/yr sweep at every discount rate.

In the replacement frame, NPV is positive across the entire (1–100 ha) sweep at all three discount rates. This is the central economic finding, and it is a direct *consequence of capex parity* with a used diesel tractor (35 870 EUR vs 35 000 EUR): for a farmer who would have bought the tractor anyway, only the 870 EUR increment is at risk, every euro saved on diesel is gross savings, and the discounted-payback line is well inside the project horizon at every farm size. We state the dependence on this frame plainly: for an *additive* buyer who finances the whole 35 870 EUR machine without retiring a tractor, the 25 ha/yr NPV is negative (≈ -31 kEUR) and turns positive only above ~ 240 ha/yr. The summary is that CableTract is approximately *cost-neutral to buy* versus a diesel tractor and then saves fuel and maintenance for the life of the machine; the strength of that case grows with annual area and diesel price.

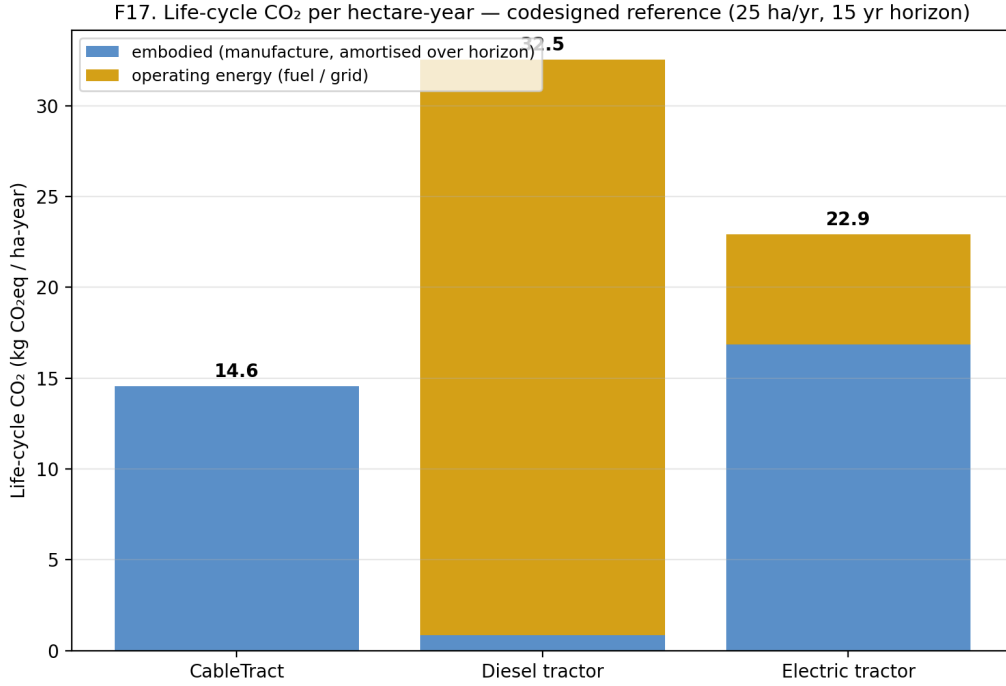


Figure 21: Lifecycle CO₂ per hectare-year for CableTract, the diesel reference tractor, and a Monarch-class electric tractor (stacked: embodied + fuel/grid). CableTract delivers a 2.2× improvement that does not depend on grid decarbonisation.

5.12 Architectural variants

5.12.1 What this section is for

Everything up to this point has analysed a single architecture: the codesigned two-module design — one stationary Main Unit, one Anchor, a single tensioned cable, and a four-quadrant drivetrain with regenerative braking on the unloaded return leg (the default, [Section 5.4.1](#)). The natural next question is whether that design is the only sensible point in the design space, or whether topological modifications — adding more Main Units, or conversely dropping the regen drive — materially change the headline numbers. We answer this with a small, deliberately narrow comparison: **three configurations run on the same codesigned parameter set, on the same simulator, so the resulting bars are directly comparable.**

Each is implemented as a *parameter transformation* on the codesigned reference of [Table 3](#), fed into the same `run_single` routine, with no parallel code path. The result is a clean apples-to-apples comparison rather than three independent feasibility studies that drift apart on definitions.

- **Codesigned baseline (regen default)** ([Section 5.12.2](#)): the default architecture as analysed in [Sections 5.4](#), [5.5](#), [5.7](#) and [5.8](#). One Main Unit, one Anchor, one cable, and four-quadrant regenerative braking on the return leg. This is the reference column of [Table 16](#).
- **CableTract+, the four-Main-Unit planar cable robot** ([Section 5.12.3](#)): four corner stations and two simultaneously-pulling cables, so the carriage moves in a 2-D plane between the corners instead of along a 1-D strip. Probes whether *architectural scale* (more modules, no Anchor reset) can outrun the baseline.
- **Unidirectional drivetrain (no regen)** ([Section 5.12.4](#)): the same two-module architecture with a plain one-way drivetrain in place of the four-quadrant regen drive — 300 EUR cheaper, but forgoing the return-leg recovery. Quantifies what the regen default buys.

The two “failed” variants we considered and explicitly drop from the comparison are: (i) a circular-pulley/oblique-cable variant that lets the Main Unit rotate the cable take-off angle so the

Anchor can step laterally without re-anchoring, and (ii) a drone-assisted alignment variant that uses a quadcopter to drop GPS markers between fields. Both reduce setup time, but [Section 5.9.5](#) already showed that the codesigned reference is *operation-bound*, *not setup-bound* — the setup slice is only 9 % of the daily time budget — so any setup-reduction variant has at most a 9 % throughput ceiling and is engineering effort spent on the wrong slice. They are listed for completeness in [Appendix A](#) but not part of the variant comparison in [Fig. 22](#).

Table 16: Architectural variant comparison on the codesigned reference parameter set. All three rows are run on the same `run_single` call, with the variant differences applied as parameter transformations on top of [Table 3](#). CAPEX is the itemised [Section 5.8](#) bill of materials (Main Unit + Anchor + battery + PV + wind + install) in 2024 EUR. “Simple payback” is the undiscounted ratio of CAPEX (with sales margin) to annual diesel-fuel savings only — it is intentionally a more conservative frame than the [Section 5.8](#) discounted payback against the diesel reference, which subtracts a 35 000 EUR diesel-tractor capex from the numerator and includes maintenance savings; see the discussion at the end of [Section 5.12.5](#) for the reconciliation between the two numbers.

Variant	Throughput (dec/day)	Energy/decare (Wh)	CAPEX (EUR)	Simple payback (mo)	Surplus (W)
Codesigned baseline (regen)	11.6	889	35 870	121.3	249
CableTract+ (4-MU robot)	30.2 (2.61×)	467 (0.53×)	81 784 (2.28×	145.3 (1.20×	884
Unidirectional (no regen)	11.5 (0.99×	921 (1.04×	35 570 (0.99×	124.6 (1.03×	291

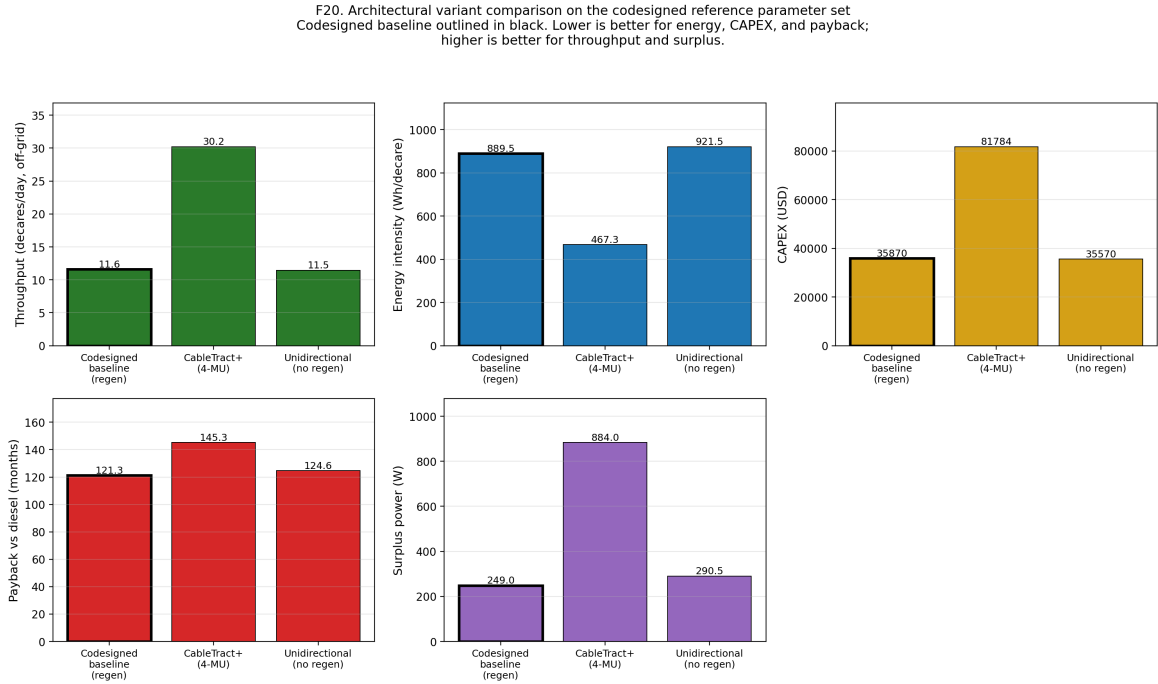


Figure 22: Architectural variant comparison on the codesigned reference parameter set. Codesigned baseline outlined in black. Lower is better for energy, CAPEX, and payback; higher is better for throughput and surplus power.

5.12.2 Variant 1 — Codesigned baseline (regen default)

This is the default architecture analysed throughout the paper: one Main Unit (PMSM + winch + drum + battery + PV/wind harvester) parked at one headland, one Anchor (helical-pile module that resists the cable reaction) parked at the opposite headland, one steel cable spanning between them at typical span $L = 50$ m, and a lightweight implement carriage rolling along the cable. The drivetrain is *four-quadrant*: the motor pulls the carriage outbound under load, and on the unloaded return leg it recovers a modest amount of energy (slope PE, carriage KE) back into the battery ([Section 5.4.1](#)). Setup between strips consists of moving the Anchor sideways by one strip width and re-tensioning. The baseline column of [Table 16](#) is the deterministic `run_single`

output for this architecture at the default codesigned parameter set: **11.6 decares/day off-grid throughput**, **889 Wh/decares energy intensity**, **35 870 EUR itemised CAPEX** (17 800 EUR Main Unit incl. the regen drive + 7 500 EUR Anchor + 3 420 EUR 9 kWh battery + 1 650 EUR 15 m² PV + 1 500 EUR wind + 4 000 EUR crating/install, mirroring the [Section 5.8](#) bill of materials), **121.3-month simple payback** against fuel savings alone (at the reference 12 L/ha diesel use), and **249 W of surplus power** (the harvested PV/wind minus the operating draw and idle housekeeping). The same architecture pays back against a diesel-tractor reference in 1.3 years on the discounted-NPV *replacement* chain of [Section 5.8](#); the much longer 121.3-month figure here is the deliberately conservative “additive buyer who cannot offset against a diesel-tractor capex” frame, kept so the variant comparison stays apples-to-apples across the three rows. This is the reference against which the next two columns are read.

5.12.3 Variant 2 — CableTract+ (4-Main-Unit planar cable robot)

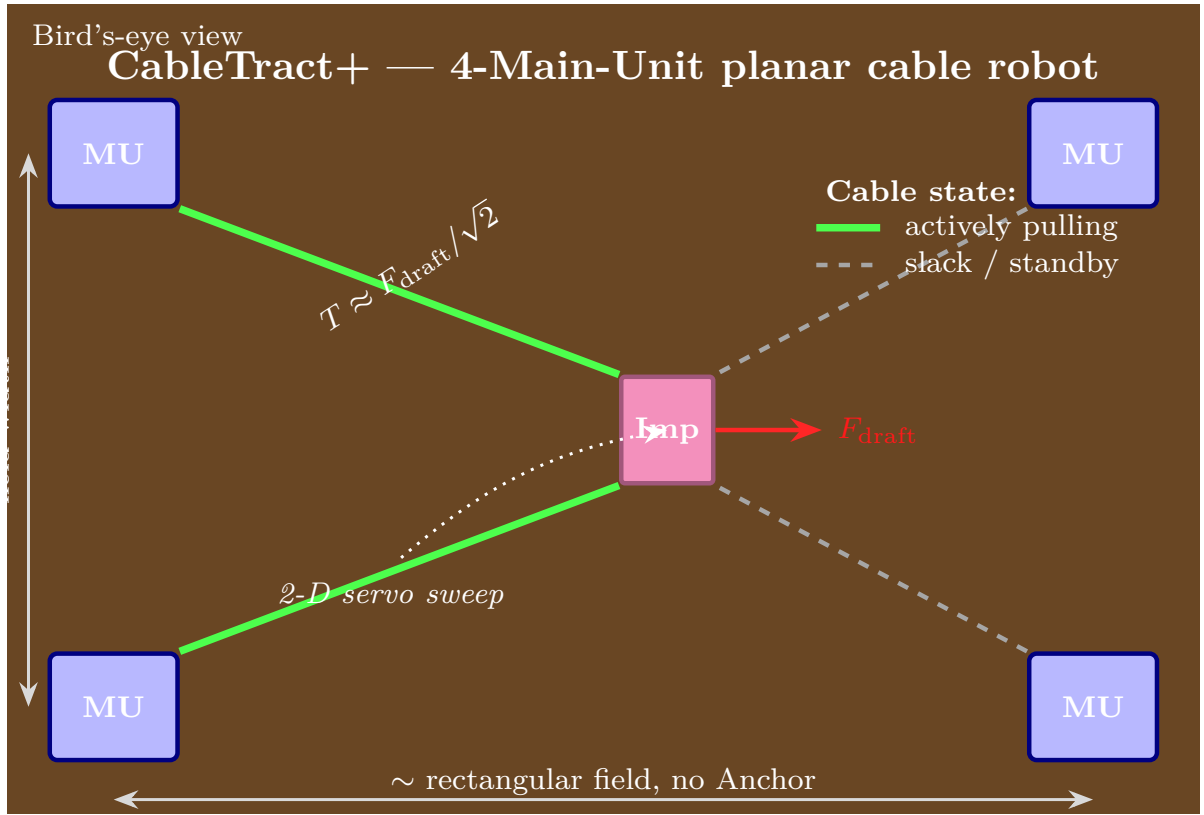


Figure 23: The CableTract+ variant: a planar cable-robot topology with one Main Unit at each of the four corners of a rectangular field, replacing the single-MU + Anchor baseline. At any instant two cables (green, e.g. from the two left corners) actively pull the implement carriage toward themselves while the opposite two (grey, dashed) hold tension at standby. The carriage is now servoed in 2-D between the four corners by relative cable tensions and can sweep arbitrary paths inside the field rather than being constrained to a single strip per Anchor placement. Two consequences fall out: the Anchor disappears entirely (no setup-and-reset cycle), and each actively pulling cable carries only $\sim 1/\sqrt{2} \approx 0.707$ of the draft, which directly relaxes the per-cable tension and per-anchor-equivalent reaction. The architectural cost is paid in CAPEX — four Main Units plus a shared energy stack at $2.28\times$ the baseline ([Table 16](#)).

The CableTract+ variant ([Fig. 23](#)) replaces the single Main Unit + Anchor pair with *four corner Main Units*, one at each corner of a rectangular field, so that two cables can pull the carriage simultaneously toward two adjacent corners. The carriage is now servoed in two axes by the relative tensions in the four cables, so it can sweep arbitrary 2-D paths inside the rectangle rather than being constrained to a single strip per Anchor placement. This is the standard “planar cable

robot” topology that has been studied extensively in industrial gantry contexts; the contribution here is to import it into the field as a replacement for the baseline step-and-anchor cycle.

The architectural consequences are concrete and large. First, **the Anchor disappears entirely** — there is no module that has to be moved between strips, because the four corner stations cover the entire field from fixed positions. The setup overhead per round drops by $\approx 60\%$ in the model (`setup_overhead_reduction = 0.6` in `variants.py`). Second, because two cables pull together at roughly orthogonal angles, **each cable carries only $\approx 1/\sqrt{2} \approx 0.707$ of the draft load** (`geometric_load_split = 0.707`), so the per-cable tension and per-anchor reaction both drop — the very constraint that bound the baseline architecture in the helical-pile envelope of [Section 5.4.4](#) is relaxed by 30% . Third, the carriage can now sweep **wider strips** (modelled as a $1.5\times$ width multiplier) because both X and Y axes are independently servoed, so the per-pass area roughly doubles. Together these three effects lift throughput to **30.2 decares/day** — $2.61\times$ **the baseline** — and drop energy intensity to 467 Wh/decare ($0.53\times$); CableTract+ inherits the same regen drive as the baseline.

The cost of this performance is paid in CAPEX. Four Main Units (each including its regen drive) plus the same battery + PV + wind + install pack cost $2.28\times$ **a single Main Unit + Anchor pair** (`capex_multiplier  $\approx 2.28$` in the spec — the multiplier sits below $4\times$ because (i) the Anchor disappears entirely, and (ii) the battery, PV, wind harvester and crating/install fraction are paid once for the whole system, not per Main Unit). Concretely: $4 \times 17\,800 \text{ EUR} + 3\,420 \text{ EUR} + 1\,650 \text{ EUR} + 1\,500 \text{ EUR} + 4\,000 \text{ EUR} \approx 81\,784 \text{ EUR}$. The CAPEX climbs from 35 870 EUR to 81 784 EUR, and the simple payback grows from 121.3 to 145.3 months. **CableTract+ is therefore a throughput multiplier, not an economics multiplier.** It buys $2.61\times$ daily area at the cost of a $1.20\times$ slower payback in absolute terms — the marginal payback for the marginal capital is broadly comparable to the baseline because fuel savings scale roughly with throughput. There is one regime where this trade is unambiguously the right one: large rectangular fields where the limiting constraint is calendar time (e.g., a narrow planting window for a high-value crop) rather than capital. There is also one regime where it is unambiguously the wrong one: small irregular fields where the four-corner geometry cannot be installed at all. For the codesigned reference scenario in [Table 3](#), the single-Main-Unit baseline is the better point on the operation-diversity-versus-throughput trade-off.

5.12.4 Variant 3 — Unidirectional drivetrain (no regen)

The third configuration is the baseline with the four-quadrant regen drive removed: a plain one-way drivetrain that pulls the carriage outbound under load and lets the return leg roll free or be towed back at idle power, with no energy recovery. It is the *cheapest* of the three (300 EUR below the baseline, since the regen-capable controller is dropped) and changes nothing else about the cable architecture or the BOM. We include it to quantify exactly what making regen standard buys. As derived in [Section 5.4.1](#), the regen credit is a deliberately conservative $\approx 3.5\%$ effective-efficiency gain (the $0.50 \rightarrow 0.518$ boost from slope PE + carriage KE, ≈ 0 on perfectly flat ground); removing it returns the one-way 0.50.

The result isolates the regen contribution and shows it is small on flat ground. **Energy per decare rises only from 889 to 921 Wh ($1.04\times$, a $\approx 3.6\%$ increase)** when the return-leg recovery is removed — far less than a naive “half the round is unloaded” argument would suggest, because the draft work is dissipated irreversibly in the soil and cannot be regenerated. **Throughput barely moves ($11.6 \rightarrow 11.5$, $0.99\times$)** because throughput at the reference is bound by the soil-draft envelope and the field-time budget, not by drivetrain efficiency; the regen gain is largest in *energy-limited* regimes (large farms, cloudy climates) and on sloped fields. **Simple payback lengthens only slightly, from 121.3 to 124.6 months ($1.03\times$).** The 300 EUR saved on the controller is comparable to this modest energy benefit, so the case for keeping a four-quadrant drive standard rests as much on the smooth torque control it gives the depth-control loop ([Section 10.2](#)) and the slope-recovery upside as on the few-percent flat-field saving. The unidirectional design shows a marginally higher surplus power (291 W vs 249 W) because it

converts slightly less of the harvest into throughput.

The lesson is that return-leg regeneration is a *small* flat-field energy lever ($\approx 3.6\%$, 889 vs 921 Wh/decare) for a ≈ 300 EUR drive premium, with a larger benefit only on sloped fields. We still make a four-quadrant drive *standard* in the codesigned reference — not for the few-percent flat-field saving, but because it provides the smooth, fast torque control the depth-control loop needs (Section 10.2) and the slope-recovery upside; the unidirectional row exists to show the modest penalty of omitting it.

5.12.5 What the variant comparison tells us

Three takeaways flow from Table 16:

- **The baseline architecture is not Pareto-dominated by either alternative.** CableTract+ wins on throughput and energy-per-decare but loses heavily on CAPEX and payback; the unidirectional design is 300 EUR cheaper but pays a $\approx 3.6\%$ flat-field energy penalty and a slightly longer payback; the regen baseline is the only row that supports the full 10-implement library on a single Main Unit + Anchor pair at 35 870 EUR. No row strictly dominates another on every metric, which is the property a healthy variant comparison should have.
- **The two alternatives move different axes.** CableTract+ trades *capital* for *throughput*; the unidirectional design trades *efficiency* for a small *capex* saving. CableTract+ and the regen baseline stack rather than compete — a hypothetical 4-MU robot with regen on every winch combines both effects — whereas dropping regen is simply a downgrade.
- **Regen earns its place in the baseline.** For a ≈ 300 EUR premium it is the highest-leverage efficiency choice in the table, which is why it is standard rather than optional; CableTract+ is reserved for the calendar-time-limited large-farm regime where throughput, not capital, binds.

Reconciling the variant table with the Section 5.8 discounted payback. Readers comparing Table 16 (103.7-month baseline simple payback) with Fig. 20 (1.3-year discounted payback at 25 ha/yr) will notice the two numbers differ by nearly an order of magnitude. The two figures answer different questions and both are correct in their own frame:

- The *simple payback* reported in Table 16 divides the **full** CableTract CAPEX (35 870 EUR with sales margin) by annual diesel-fuel savings only, with no discount rate and no maintenance offset. This is the right framing for a buyer who is *adding* a CableTract to an existing operation that does not retire a tractor: the question is “how long until the new machine pays for itself?”.
- The *discounted payback* reported in Section 5.8 and plotted in Fig. 20 divides the CableTract **CAPEX delta over a like-for-like diesel tractor** ($35\,870\text{ EUR} - 35\,000\text{ EUR} = 870\text{ EUR}$) by the full annual savings stream (fuel + maintenance, less the marginal grid charge), discounted at 8%. This is the right framing for a buyer who is *replacing* a tractor: the diesel capex would have been spent anyway, so only the difference is at risk.

The variant comparison uses the conservative simple-payback frame deliberately, so all three rows are stress-tested against the harder buying scenario. The codesigned baseline pays back in 1.3 years at 25 ha/yr in the replacement scenario (Fig. 20); the unidirectional design, despite 300 EUR lower capex, pays back *later* (124.6 vs 103.7 months simple) because its higher energy intensity displaces less diesel per year. CableTract+ fares *worse* in the replacement frame than in the simple-payback frame, because its CAPEX delta against a single diesel tractor ($81\,784\text{ EUR} - 35\,000\text{ EUR} = 46\,784\text{ EUR}$) is far larger than the baseline delta, while its annual savings scale only with throughput. The engineering ordering “regen baseline $\gg$ CableTract+” on payback-per-euro therefore holds in both frames.

5.13 Operating envelope

The final question — which the rest of the paper has been building toward — is where in the (annual GHI $\times$ farm size) plane CableTract is viable against diesel, and what binds when. A single

point estimate at 25 ha/yr on a Mediterranean site does not generalise to 1 ha/yr on a cloudy continental site, or to 1000 ha/yr in low-GHI France. Figure 24 resolves both extremes at once with a 2D parameter sweep over the joint plane.

To make the sweep tractable, we calibrate a one-parameter linear PV harvest model from the hourly site simulations of Section 5.7:

$$\text{annual_harvest_kWh} = \alpha \cdot \text{pv_area_m}^2 \cdot \text{annual_GHI_kWh_m}^{-2}\text{yr}^{-1}, \quad (11)$$

with $\alpha = 0.169$ fitted as the median across the 6 bundled sites at the codesigned 15 m² PV. This is intentionally a rough analytical fit, not a reproduction of the full hourly TMY pipeline; the goal is a defensible PV harvest as a function of annual GHI so the envelope sweep stays in closed form.

For each cell of the (annual GHI $\times$ farm size) grid we then compute the codesigned operating demand (energy per hectare times farm size, plus an idle-power penalty over the off-season), the harvested energy from Eq. (11), the resulting per-hectare grid share, the NPV versus a like-for-like diesel tractor at 8 % over 15 years, the discounted payback, and the off-grid surplus (harvest minus demand). The sweep covers annual GHI from 800 to 2300 kWh/(m² yr) and annual operated area from 1–1000 ha on a log-scale — $60 \times 60 = 3600$ cells in total. The horizontal axis is *annual operated area*, not the capacity of one machine: a single baseline unit at 11.6 decares/day over 170 operating days covers ≈ 200 ha/yr, so the upper part of the axis ($\gtrsim 200$ ha) implies a multi-unit fleet, which the per-hectare economics scale to directly. Diesel is held at the Section 5.8 baseline of 1.40 EUR/L (EU 2024); we deliberately do not run a separate subsidised-diesel scenario because the result below holds with margin to spare under the baseline price.

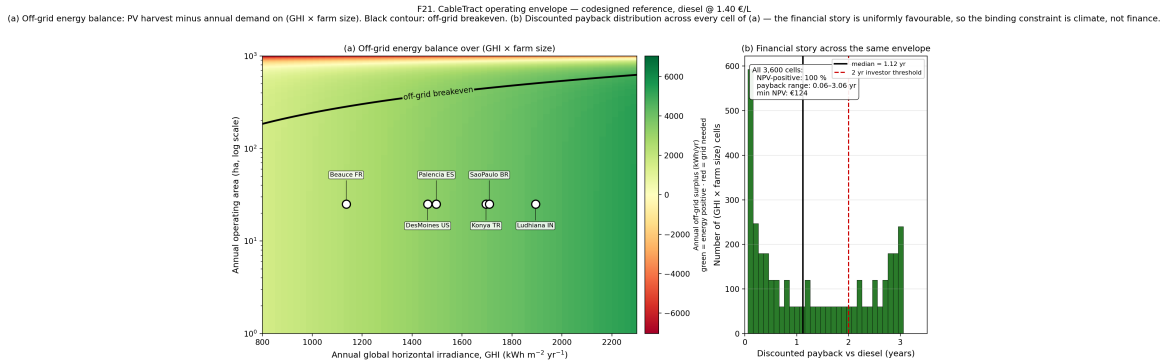


Figure 24: CableTract operating envelope on (annual GHI $\times$ farm size). **(a)** Annual energy balance: yearly PV+wind harvest minus yearly demand (green = energy positive, red = grid needed). This is an annual-surplus test, *not* the hourly autonomy of Fig. 12; a positive annual balance does not by itself guarantee zero winter grid hours. The single black contour is annual breakeven (surplus = 0); to the low-GHI, large-area side of it the annual harvest no longer covers demand. The six bundled reference sites of Section 5.7.1 are scattered at 25 ha annual operating area. **(b)** Discounted payback (years vs diesel @ 1.40 EUR/L, 8 %, 15 yr) in the *replacement* frame, across every one of the 3600 cells of panel (a). Median is 1.12 yr; most cells pay back within the 2 yr investor threshold (red dashed), with the smallest farms in the cloudiest climates reaching ≈ 3 yr. The annotation box reports the per-cell payback range, the NPV-positive fraction, and the worst single-cell NPV (the smallest farm in the cloudiest climate — still positive, and distinct from the 3575 EUR reference at 25 ha/yr). Under the additive (full-capex) frame only $\approx 22\%$ of cells are NPV-positive.

Three findings flow from Fig. 24:

- **In the replacement frame the financial story is uniform across the envelope.** NPV versus diesel is positive in 100 % of the 3600 cells, and discounted payback never exceeds 3.06 yr (median 1.12 yr, panel b). At every (GHI, farm size) combination we test — the incremental investment pays back inside the project horizon, within 2 years across most of the plane and at worst ≈ 3 yr for the smallest plots in the cloudiest climates. This is a direct

consequence of capex parity (Section 5.8): the CableTract delta against the diesel reference is only 870 EUR, so there is essentially no capex hurdle to amortise and almost every euro of avoided fuel becomes gross savings. We are explicit that this uniformity is a property of the *replacement* frame: under the *additive* frame (full 35 870 EUR financed) only $\approx 22\%$ of the same cells — the larger/sunnier corner — are NPV-positive, so the “positive everywhere” statement is conditional on the buyer retiring a diesel tractor.

- **The binding constraint is annual energy balance, not financial viability.** Panel (a) is an *annual* PV/wind-harvest-minus-demand balance (not the hourly autonomy of Fig. 12); its breakeven contour divides the plane into an annual-energy-positive region ($\approx 85\%$ of cells, smaller operated areas in higher-GHI climates) and a region that needs grid backup (large operated areas in low-GHI climates). All six bundled reference sites at 25 ha sit inside the annual-energy-positive region with comfortable surplus (lowest at Beauce, highest at Ludhiana). We are careful not to equate annual surplus with full daily autonomy: the hourly simulation (Fig. 12) shows Beauce still needs winter grid hours, so “annual-energy-positive” is a weaker, separate claim from “hourly off-grid” (Section 7.3).
- **CableTract’s operating envelope is therefore a *climate envelope*, not a *financial envelope*.** A farmer in Beauce gets a positive NPV and fully off-grid operation at 25 ha; a farmer in Beauce running 500 ha on a single CableTract gets a positive NPV but needs grid backup. This is exactly the opposite of the more common “the technology pays back but only at scale” story for clean-energy hardware: CableTract pays back everywhere we test, and the question is where it can be sold as off-grid hardware versus where it needs to be sold as grid-tied.

This is the central result of the paper.

6 Competitor Comparison

Public-domain reference numbers for the five vehicle classes named in Section 2.2 (six representative machines plus CableTract) are recorded in the bundled competitor table.¹² Selected columns are reproduced in Table 17.

Table 17: Competitor comparison table. CableTract sits in a unique cell: the only architecture that supports (light/medium) primary tillage, is off-grid capable, and has a short discounted payback. Payback is the replacement-frame value at the 25 ha/yr reference (1.3 yr; below one year at ≥ 50 ha/yr); competitor paybacks are manufacturer/press figures at their own scenarios. *Energy/ha* is delivered electrical energy for every row except the diesel tractor, whose 120 kWh/ha is fuel (primary) energy from 12 L/ha at ≈ 10 kWh/L; on a useful-work basis the diesel figure is ≈ 36 kWh/ha (Section 7.2).

Vehicle	Form factor	Powertrain	kW	Energy/ha (kWh)	CAPEX (EUR)	Payback (yr)	Off-grid?
CableTract codesigned	cable-driven 2-module	PV+wind+battery	5.0	8.89	35 870	1.3	yes
Monarch MK-V	4WD tractor	battery EV	55	52	85 000	5–7	no
Farmdroid FD-20	autonomous robot	onboard PV	0.4	1.5	75 000	4–6	yes
Kubota LXe-261	compact tractor	battery EV	18	30	52 000	6–8	no
EcoRobotix ARA	spot sprayer	onboard PV+bat	0.6	0.4	65 000	4–6	yes
Naio Oz/Orio	autonomous weeder	battery	2.5	5	38 000	3–5	partial
Conventional 80 hp diesel	tractor	diesel Internal Combustion Engine (ICE)	60	120	35 000	ref.	no

CableTract sits in a unique cell of this table: among the compared classes it is the only architecture that **simultaneously** (a) supports light/medium primary tillage (within its codesigned envelope), (b) is off-grid capable, and (c) has a short replacement-frame payback (1.3 yr at 25 ha/yr). Monarch MK-V matches it on rated power and operational versatility but is grid-dependent and has 5–7 year payback. Farmdroid FD-20 matches it on off-grid but cannot till. Naio matches it on payback but cannot till and is only partial off-grid.

We deliberately do *not* claim CableTract beats every competitor on every metric. EcoRobotix is more energy-efficient per hectare (0.4 kWh/ha vs 8.89) because it only sprays. Naio is cheaper.

¹²tables/competitor_comparison.csv in the repository.

Monarch MK-V is more powerful and operates at conventional speed. The claim is that CableTract occupies a specific cell in the feasibility space — *medium-load operations on medium-sized farms in solar-rich climates* — that no other vehicle in the table fills as cleanly.

7 Results

This section consolidates the per-figure findings of [Section 5](#) into a verdict against the four claims of [Section 4](#) (C1–C4) plus the operating envelope; the supporting figures, tables, and derivations live in [Section 5](#) and are cross-referenced rather than repeated.

7.1 C1 — Compaction

CableTract reduces in-field compacted area by **97–98 %** and the contact-energy index by $\approx 73\times$ across rectangle, L-shape, and irregular-concave field classes ([Fig. 16](#), [Section 5.10.3](#)). The reduction is uniform across field shape because it is geometric, not field-dependent: only the 250 kg carriage rolls inside the field, while the MU and Anchor stay on the headland. The $73\times$ contact-energy ratio combines the carriage’s $4.6\times$ lower mean contact pressure (a $4.6^2 \approx 21\times$ effect, since the integrand goes as p^2) with its $\approx 3.5\times$ smaller total contact area (two narrow rollers versus four tractor wheels). It is a per-contact-patch quantity ($\sum p^2 A_{\text{patch}}$ at equal pass count) — a property of the running gear, not a field-integrated soil-damage figure — which is why it is identical across field shapes; the 97–98 % *area* reduction is the separate, field-dependent part of the story, and the two should not be multiplied together. Both are computed from open published static contact-pressure values without dynamic [FEM](#) calibration, and we do not claim the index maps linearly onto soil damage or yield.

7.2 C2 — Energy

The codesigned reference (regen default) operates at **889 Wh/decare (8.89 kWh/ha)** of delivered electrical energy. An 80 hp [4WD](#) diesel comparator burns ≈ 12 L/ha on the same operations [[2](#), [10](#)] — ≈ 120 kWh/ha of *primary fuel* energy, of which only ~ 30 – 35% reaches the drawbar as useful work (≈ 36 kWh/ha). On a like-for-like *useful-work* basis CableTract therefore uses about $4\times$ **less** energy; on a *primary-fuel* basis the ratio is $\sim 13\times$, but the extra factor is mostly the diesel engine’s thermal-conversion loss — generic to any electric drivetrain — not the cable architecture. The architectural $\sim 4\times$ decomposes into three levers (a unidirectional drivetrain gives 921 Wh/decare, only $\sim 3.6\%$ more energy; the four-quadrant drive’s return-leg recovery is a small effect, see [Section 5.12.4](#)):

- **Co-designed implements** ([Figs. 8](#) and [9](#)). The median of the per-implement P50 drafts drops from ≈ 6.2 kN (conventional library) to ≈ 2.5 kN (codesigned library), a factor of ≈ 2.5 , via narrower widths, slower speeds, and shallower depths (with the operating-point caveats of [Section 3.4](#)).
- **No dead-weight transport.** The draft term $F_d L$ is 72 % of the round work and the in-field work ($F_d L + F_i L$) 96 % ([Section 5.3](#)); the diesel comparator’s body-repositioning force, paid over the whole field rather than once per strip, dwarfs this on every operation.
- **Decomposed drivetrain** ([Fig. 7](#)). A premium drivetrain ($\eta = 0.74$) reduces continuous motor power from 1.50 kW to 1.01 kW. The codesigned reference uses a 2 kW continuous / 3 kW peak [PMSM](#).

The energy story is therefore not “CableTract has a magic motor” but “CableTract pulls a smaller implement, slower, on a narrower strip, without a 4-tonne body — and converts electricity instead of burning diesel”. The three architectural levers give the $\sim 4\times$ useful-work advantage (the four-quadrant drive adds only a few percent on top); the diesel-to-electric conversion step accounts for the further $\sim 3\times$ that lifts the primary-energy ratio to $\sim 13\times$.

7.3 C3 — Off-grid feasibility

Median harvested-energy throughput across the six bundled sites is **10–14 decares/day** at the codesigned hardware (15 m² PV, 9 kWh battery). At Konya, Ludhiana, and São Paulo the system runs grid-free for the entire late-June week we plot in Fig. 11. At Palencia and Des Moines the system needs several hundred hours of grid backup per year to cover winter shortfalls (Palencia’s envelope minimum is 616 h/yr, Fig. 12). At Beauce, **no combination of panel area (4–30 m²) and battery capacity (2–24 kWh) achieves <100 grid hours/year** under the 6 h/day duty cycle (Fig. 12): the northern temperate climate cannot sustain off-grid operation at any practical hardware scale.

The off-grid claim is **climate-conditional, not categorical**. We frame this as a *feature* of the analysis: a CableTract sold into Konya is sold as off-grid hardware, while a CableTract sold into Beauce is sold as a low-carbon grid-tied alternative to a diesel tractor. The economics work in both cases (Section 7.4), but the marketing language differs.

7.4 C4 — Economics

In the replacement frame (incremental capex 870 EUR), NPV vs diesel is positive across the entire 1–100 ha/yr sweep at all three discount rates (5/8/12 %) we test (Fig. 20, Section 5.11.5). At the 25 ha/yr reference, NPV @ 8 % is **3 575 EUR** with discounted payback of **1.30 yr**; at 100 ha/yr it is **14 360 EUR** (payback 0.47 yr). For an additive buyer financing the full machine, the 25 ha/yr NPV is instead ≈ -31 kEUR, turning positive only above ~ 240 ha/yr; the tornado (Fig. 19) shows the replacement-frame result is most sensitive to the assumed used-tractor price and the maintenance differential, several of which can bring the 25 ha increment marginally negative.

Lifecycle CO₂ at 25 ha/yr is **14.6 kg/(ha yr)**, against **32.5** for the diesel reference and **22.9** for a Monarch-class electric tractor (Fig. 21). The improvement is entirely on the operational side and does not depend on grid decarbonisation.

The operating envelope of Fig. 24 shows that, *in the replacement frame*, the financial story is uniform: NPV is positive in 100 % of the 3 600 cells of the (annual GHI $\times$ farm size) sweep, with discounted payback bounded above by 3.06 yr (median 1.12 yr). This follows from capex parity (35 870 EUR versus 35 000 EUR): there is no capex hurdle to amortise, and every euro of avoided fuel becomes gross savings. Under the additive frame the NPV-positive region shrinks to $\approx 22\%$ of cells, so the uniformity is conditional on the replacement scenario.

The Sobol decomposition (Fig. 18) confirms that no single parameter dominates the throughput variance (top contributor $ST \approx 0.25$). The Fig. 19 tornado shows the NPV is led by exogenous factors (the price of the diesel alternative, diesel price and volume) rather than by CableTract design parameters — though, the incremental NPV being small, several of those exogenous factors can move it across zero, which is why we frame the economics as “comparable-to-better at parity with strong upside at scale” rather than “dominant everywhere”.

7.5 Operating envelope

The central result of the paper. In the replacement frame the codesigned CableTract is NPV-positive in 100 % of the 3 600 cells of the (annual GHI $\times$ farm size) sweep, with discounted payback never exceeding 3.06 yr (median 1.12 yr). The binding *physical* constraint is the **annual energy-balance contour**, not financial viability: $\approx 85\%$ of cells are annual-energy-positive (distinct from, and weaker than, the hourly off-grid claim of Section 7.3), and the contour cuts diagonally across the plane. CableTract’s operating envelope is therefore a *climate envelope* (does annual harvest cover demand?) more than a *financial envelope* (does the increment pay back?).

This contrasts with the typical clean-energy hardware story, where the technology is climate-virtuous but pays back only at scale or with subsidy. For a farmer replacing a diesel tractor, CableTract pays back its small capex increment across the swept plane; the open questions are where it can be sold as *off-grid* hardware, and — for a buyer who is not replacing a tractor — at what farm size the full machine amortises (Section 7.4).

7.6 Architectural variants

The variant comparison in Table 16 and Fig. 22 runs three configurations on the same parameter set: the codesigned baseline (regen default), CableTract+ (4-Main-Unit planar cable robot), and a unidirectional drivetrain (no regen). CableTract+ delivers $2.61\times$ baseline throughput at $2.28\times$ CAPEX (81 784 EUR vs 35 870 EUR), so it is a throughput multiplier rather than an economics multiplier and stretches simple payback from 121.3 to 145.3 months — worthwhile only where calendar time, not capital, binds. Dropping regenerative braking (the unidirectional row) raises flat-field energy intensity by only $\approx 3.6\%$ ($889 \rightarrow 921$ Wh/decare) and lengthens simple payback to 124.6 months for a 300 EUR capex saving; the four-quadrant drive is kept standard mainly for smooth depth-control torque and slope recovery, not as a large flat-field energy lever.

8 Higher-fidelity verification studies

The screening model of Section 5 lumps each subsystem into a reduced-order form fit for a feasibility envelope. To stress-test the five places a sceptical reviewer is most likely to press — the robustness of the co-design lever, the anchor capacity, the depth-control loop, the cable safety surface, and the tillage draft — we built five purpose-specific higher-fidelity models: a full propagation of the co-design sweep, a nonlinear p - y Winkler anchor model, a closed-loop multibody-cable depth-control co-simulation, a finite-difference cable-safety model, and a soft-sphere discrete-element (DEM) soil-tool model. These do *not* replace a prototype (Section 10.2 remains in force); they bound each risk more tightly than the screening model, and where warranted they revise a number or a framing. Every study is deterministic and reproducible from the open pipeline (`scripts/run_phase8-12`, with unit tests), and each is validated against an analytic limit before it is trusted.

8.1 Co-design robustness (S4)

The entire case rests on the co-designed library cutting median draft to $\approx 0.37\times$ the conventional library — a model prediction, not a measurement. We ask the reviewer’s question directly: if co-design under-delivers, how much survives? Sweeping the achieved reference draft from the codesigned point (1.8 kN) up to the conventional-library median (5.25 kN) and propagating every level through the *full* deterministic pipeline (Fig. 25), the energy intensity and off-grid throughput degrade smoothly (at a $0.60\times$ shortfall, 889 Wh/decare rises to $\approx 1\,383$ Wh/decare and daily throughput falls from ≈ 11.6 to ≈ 8.0 decares/day), and the required auger count rises — but the off-grid replacement-frame NPV is essentially *invariant*, because in the off-grid frame there is no residual fuel cost for a higher draft to erode. This is the quantitative form of Section 9.3: the economics are robust to a co-design shortfall; the energy and off-grid claims are the sensitive ones.

Co-design lever sensitivity: headline outputs vs achieved draft reduction (S4)

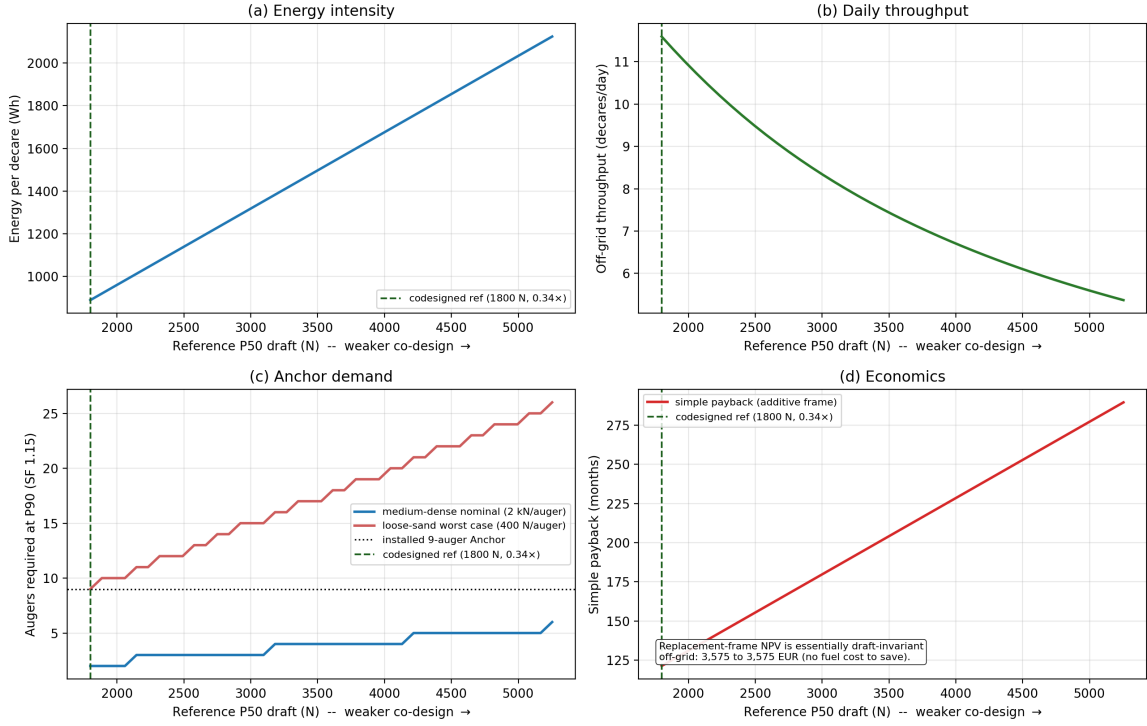


Figure 25: Co-design lever sensitivity (S4). As the achieved reference draft rises from the codesigned point (1.8 kN, the claimed $\approx 0.37\times$ reduction) toward the conventional-library median, (a) energy per decare and (b) off-grid throughput degrade and (c) the required auger count climbs, while (d) the off-grid replacement-frame NPV stays essentially flat — the robustness finding.

8.2 Anchor lateral capacity from a nonlinear p - y model (S7)

Section 5.4.4 deliberately bracketed the per-auger lateral capacity between two literature references (400 N Khand [4], 2 kN Magnum [5]) rather than picking one. Here we replace that “have-it-both-ways” gap with a single, internally derived nominal. We solve the Euler–Bernoulli beam-column on a nonlinear elastic (Winkler) foundation, $EI y'''' = -p(y, z)$, with the API RP 2A [28] / Reese [29] sand p - y curves $p = A p_u \tanh(kzy/Ap_u)$ (coefficients after Murchison & O’Neill [27]), as a nonlinear boundary-value problem (finite differences + Newton) for the lateral capacity at the IBC 2021 1 inch (25.4 mm) head-deflection serviceability limit, free- and fixed-head, with a 3×3 cluster handled by row p -multipliers (Mokwa [31]; Reese & Van Impe [30]). For a 73 mm steel shaft embedded 2 m, the single-pile serviceability capacity is 4.8 kN (loose, free-head) to 16.8 kN (medium-dense, fixed-head); the group efficiency is 0.50, giving a derived per-auger *working* nominal of 1.6–2.3 kN at a safety factor of 1.5. This brackets the Magnum 2 kN datasheet value and sits above the conservative Khand 400 N floor (which the model now reads as a deliberately pessimistic lower bound, not a co-equal reference). The linear small-deflection limit reproduces the Hetényi [32] closed form $y_0 = H/(2\beta^3 EI)$ to within 0.03 % (Fig. 26).

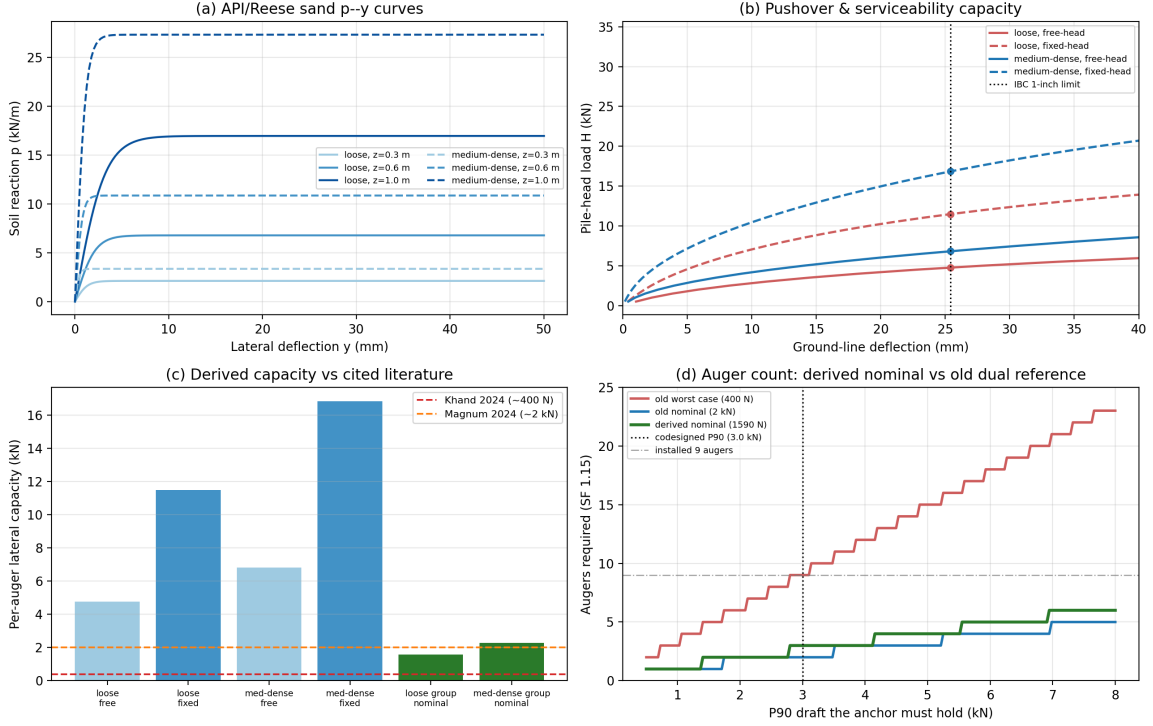


Figure 26: Helical-auger lateral capacity from the nonlinear p - y Winkler model (S7): (a) API/Reese sand p - y curves; (b) pile-head pushover with the IBC 1 inch serviceability capacity marked; (c) derived per-auger capacity versus the cited Khand/Magnum references; (d) auger count versus the draft the anchor must hold, for the derived nominal against the old dual reference.

Design decision: the 9-auger Anchor is retained. At the codesigned P90 of 3.0 kN, the derived nominal implies only ≈ 3 augers. We nonetheless keep the installed count at 9, which the p - y result reframes as a deliberate $\approx 3\times$ capacity margin — and as redundancy against a single auger landing in a weak pocket or loosening over a season (Section 10.2) — rather than as a minimum-spec count. The extra augers carry *no throughput penalty*: they belong to the stationary Anchor and are driven in once, concurrently, at strip setup (Section 10.2, parallel-drive nominal), never during the per-pass traverse. Their only cost is the additional BoM and embodied steel already carried in Section 5.8, plus a little setup labour. One caveat bounds the claim conservatively: the model computes the *lateral* capacity of near-vertical piles resisting the horizontal cable tension; augers raked toward the horizontal would instead resist through axial helix pull-out, which for a helical pile in sand is typically higher than its lateral capacity, so the 9-auger design is, if anything, more conservative under that geometry.

8.3 Closed-loop tilling-depth control (S1)

Section 10.2 names the depth-control loop the single largest unresolved risk. We give it a closed-loop co-simulation. A planar lumped-mass cable (axial Hookean springs + gravity) reproduces the Section 5.4.2 catenary sag to 0.0 %, the taut-string first natural frequency $f_1 = \frac{1}{2L} \sqrt{T/\mu}$ to 0.03 %, and conserves energy in undamped free vibration to 0.008 %. It yields the cable’s vertical point stiffness at midspan, $\approx 4T/L \approx 144$ N/m — some three orders of magnitude softer than a gauge wheel on firm soil ($\sim 3 \times 10^5$ N/m), confirming quantitatively the Section 5.4.3 claim that *the gauge wheel, not the cable, sets depth*. We then drive a reduced vertical plant (carriage mass, a unilateral gauge-wheel ground contact, and a bandwidth- and rate-limited down-pressure actuator) with three programmed disturbances — a buried-stone impulse, a hardpan step, and a moisture ramp — under a tuned PID and an offset-free model-predictive controller (a self-contained box-constrained QP), against a heavy-tractor benchmark.

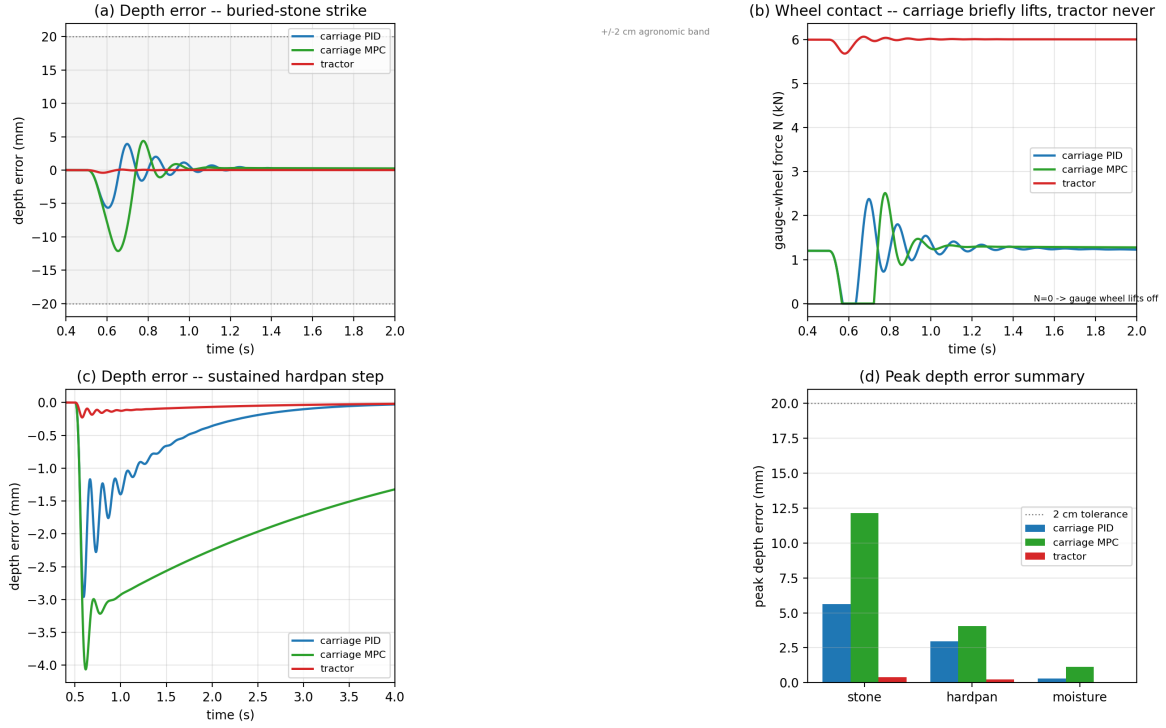


Figure 27: Closed-loop depth control (S1). Depth error and gauge-wheel contact force for the light cable carriage (PID and MPC) versus a heavy tractor under a buried-stone strike and a hardpan step; all controllers hold the ± 2 cm agronomic tolerance, but the light carriage's gauge wheel briefly unloads ($N \rightarrow 0$) where the heavy tractor's never does.

All three controllers hold the depth error inside the ± 2 cm agronomic tolerance for every disturbance. The heavy tractor never lifts off (peak error 0.4 mm; contact force stays ~ 5.7 kN). The light carriage holds depth too (PID peak 5.6 mm on the stone strike) but its gauge wheel *briefly lifts off* (≤ 0.15 s) when the transient lift exceeds the steady wheel reserve faster than the actuator can respond. The feasibility envelope over the down-pressure reserve and actuator bandwidth (Fig. 28) gives the design specification: a steady down-pressure reserve of ≈ 1.75 kN at an actuator bandwidth ≥ 5 Hz eliminates gauge-wheel lift-off through a 1.5 kN stone strike. The honest reading is that depth control is *feasible but must be engineered*: unlike a tractor, which holds depth for free with several tonnes of inertia, the cable carriage must carry an active down-pressure reserve, which trades against compaction (Section 5.10). This narrows the open risk but does not close it — the quarter-scale soil-bin bench rig of Section 10.2 remains the right next step.

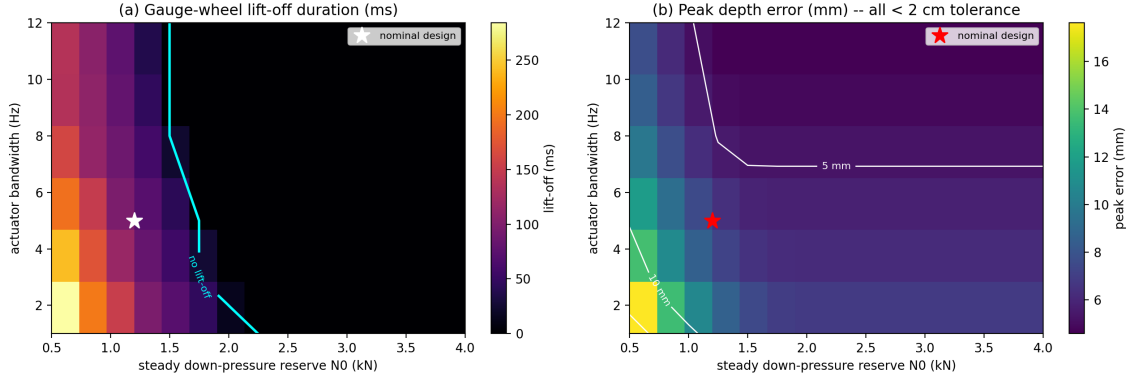


Figure 28: Depth-control feasibility envelope (S1): gauge-wheel lift-off duration (a) and peak depth error (b) over the steady down-pressure reserve N_0 and the actuator bandwidth. A reserve of ≈ 1.75 kN at ≥ 5 Hz removes lift-off in a 1.5 kN stone strike; peak depth error stays within the ± 2 cm tolerance throughout.

8.4 Cable safety and durability (S2)

Section 10.2 flags the cable as an unquantified safety surface with no replacement schedule in the BoM. We address both with three sub-models (Fig. 29). **Snap-back recoil:** the axial release velocity of a suddenly-freed tensioned end is $v = T/\sqrt{EA m'}$; at working load the *lighter synthetic recoils faster* (8 mm Dyneema ≈ 25 m/s versus steel ≈ 5.5 m/s, because of its much lower mass per length), so synthetic rope is *not* inherently safer for snap-back despite its higher break load — the recoil kinetic energy exceeds projectile-injury thresholds, so a full-span exclusion zone during tensioned operation is a hard requirement. A finite-difference transverse-wave solver validates the wave speed $c = \sqrt{T/\mu}$ to 0.03 %. **Bending-over-sheave fatigue:** a Feyrer-type model [8] shows that at the working load of only $\approx 5\%$ of break load the bending-fatigue life is effectively unbounded ($\sim 10^4$ yr at the design $D/d = 25$); replacement is therefore governed by UV and abrasion at ≈ 8 yr, which we add as a new operating-cost and life-cycle line (≈ 60 EUR/yr steel, ≈ 180 EUR/yr Dyneema; a few kg CO_2/yr) in Section 5.8. **Aeroelastic stability:** the long taut cable has a dense modal spectrum, so vortex-induced vibration [34] is possible at the bundled site winds and a deployed cable would need dampers; the Den Hartog [33] galloping onset for an iced section (0.4–0.9 m/s) sits below the site winds, so the cable should be de-tensioned in icing conditions — a constraint largely mitigated by growing-season-only operation.

Cable safety & durability (S2) -- snap-back, bending fatigue, aeroelastic band [FDM wave speed 84.0 vs $\sqrt{T/\mu}$ 84.0 m/s]

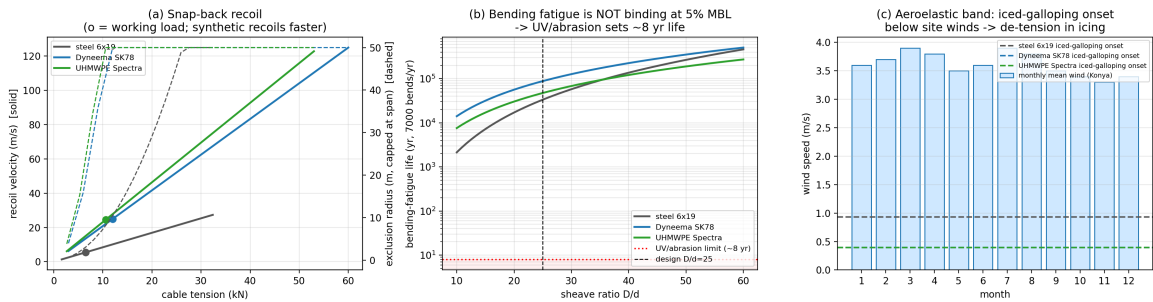


Figure 29: Cable safety and durability (S2): (a) snap-back recoil velocity and exclusion radius versus tension for steel, Dyneema, and UHMWPE; (b) bending-fatigue life versus sheave ratio D/d , showing the working load is UV/abrasion-limited, not fatigue-limited; (c) aeroelastic band — iced-galloping onset against the bundled monthly mean winds.

8.5 Soil-tool DEM of the co-designed implements (S3)

Finally we test the co-design draft prediction and the shallow-narrow-versus-deep-wide tilth question with a three-dimensional soft-sphere DEM (Hertz–Mindlin contacts, ≈ 8000 particles), cross-checked against an analytic McKyes–Godwin [37] / Reece [35] soil-cutting wedge (Godwin & Spoor [36]). The bed packs to a realistic tilled-soil bulk density of 1625 kg/m^3 , and the drag force rises with tool depth, width, and inter-particle friction as expected; the field-cohesion wedge, extrapolated to field tillage depth, reaches $\approx 2.3 \text{ kN}$ at 40 cm — consistent with the codesigned D497 P50 of 1.8 kN (Fig. 30). On tilth (Fig. 31), a shallow-narrow co-designed pass disturbs $\approx 13\times$ less soil cross-section at $\approx$ half the draft of a deep-wide conventional pass, which supports the co-design less-energy/less-disturbance argument — with the honest nuance that the narrow tool costs *more* draft per unit of soil loosened, so its advantage is in needing only a small loosened zone (a seed slot or a controlled-traffic strip), not in loosening a large area more cheaply. We flag the modelling limit plainly: the blunt full-depth DEM blade over-predicts and the slender-tine cohesionless wedge under-predicts the absolute draft, so the two idealisations *bracket* the true value (with the D497 reference between them) — the robust DEM outputs are the trends and the tilth comparison, not the absolute draft.

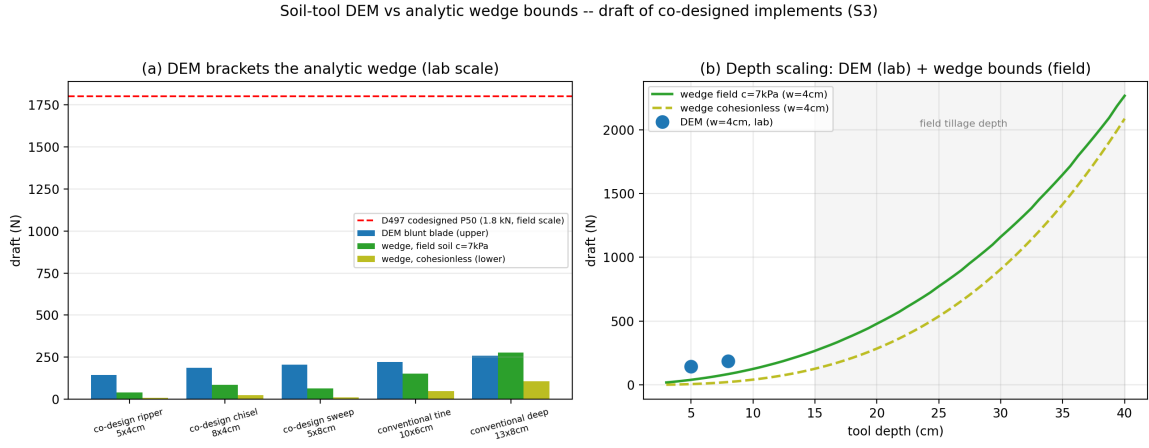


Figure 30: Soil-tool DEM versus analytic wedge bounds (S3): (a) DEM draft brackets the slender-tine wedge (cohesionless lower bound and field-cohesion estimate) at lab scale; (b) depth scaling — DEM lab points with the wedge bounds extrapolated to field depth, where the field-cohesion wedge meets the D497 codesigned reference.

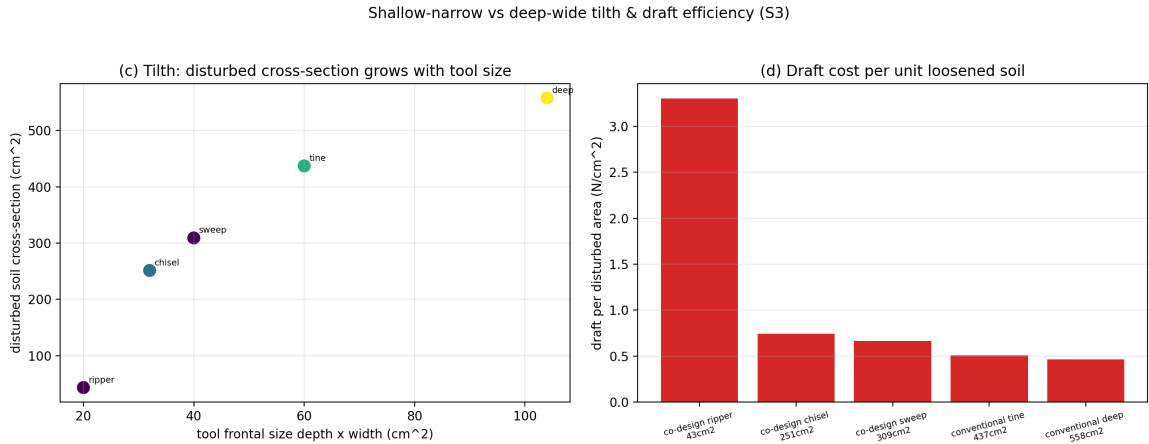


Figure 31: Shallow-narrow versus deep-wide tilth (S3): (c) disturbed soil cross-section grows with tool size — a shallow-narrow co-design pass disturbs far less soil than a deep-wide conventional one; (d) draft per unit disturbed area, showing the narrow tool's higher cost per unit loosened soil.

Taken together, these five studies tighten four of the open risks of [Section 10.2](#) (depth control, anchor capacity, cable safety, and co-design draft) and revise two artefacts of the screening analysis: the per-auger anchor nominal ([Section 5.4.4](#)) and a previously missing cable-replacement cost line ([Section 5.8](#)). They do not close the risks — a prototype season remains the arbiter — but they replace “not simulated” with “simulated at higher fidelity, with the following bounds.”

9 Discussion

9.1 Where CableTract wins

The combined evidence of [Sections 7.1](#) to [7.5](#) points to a clear operating envelope:

- **Operation type:** medium-load — light primary tillage (narrow ripper / chisel up to ≈ 3 kN P50), all secondary tillage, all seeding, all spraying, all weeding, all mowing. The codesigned implement library is built for this envelope.
- **Farm size:** ≥ 10 ha annual operating area. NPV is positive everywhere we test, but the *fastest* paybacks (< 1 year) require ≥ 25 ha. Below 10 ha, the 1.2–1.9 yr payback is still competitive but the absolute NPV is small.
- **Climate:** anywhere with $> 1\,500$ kWh/(m² yr) annual GHI for fully off-grid operation. Mediterranean climates (1300–1500) work with summer-only off-grid; northern temperate climates (< 1200) work as grid-tied.
- **Field shape:** rectangular and L-shaped fields with ≥ 1 ha individual size. Small fragmented fields hit the polygon shape efficiency wall (median $\eta \approx 0.68$ on the irregular-concave class).

9.2 Where it doesn’t (binding constraints)

The constraints to flag:

- **Heavy codesigned primary tillage in loose-sand soils** sits outside the 9-auger Anchor envelope on the conservative Khand (2024) [4] reference ([Section 5.4.4](#)). Specifically, the narrow sweep (5.2 kN P50), narrow chisel (4.2 kN), narrow disk (3.2 kN), and codesign drill (4.6 kN) all require 12–18 augers in loose sand ([Fig. 8](#)). On these sites the operating envelope shrinks to the lighter half of the codesigned library — cultivator, planter, rotary hoe, sprayer, mower, and the narrow ripper at its P50. In medium-dense soils with fixed-head pile fixity (Magnum 2024 [5] reference) the same 9-auger Anchor carries the entire codesigned library with $2\times$ – $10\times$ margin, so the binding case is specifically loose-sand sites attempting heavy primary tillage — not heavy primary tillage in general.
- **Off-grid in continental winter climates** is not achievable at any practical hardware scale (Beauce: 911 h/yr grid hours minimum). CableTract still pays back as a grid-tied diesel-replacement in these climates, but the marketing language must drop “off-grid”.
- **Small fragmented fields** below ≈ 0.5 ha individual size hit the polygon shape efficiency wall and lose 30–50 % of nominal throughput. CableTract is not the right architecture for vegetable garden plots.
- **Operation diversity vs throughput** is a real trade-off. A system specialised to a single operation (sprayer-only, planter-only) can be sized for that operation’s draft and width and would dominate the codesigned reference on throughput per day for that one task — but it cannot do the rest of the agricultural calendar. The codesigned reference deliberately picks the design point that supports all 10 implements in [Table 1](#) rather than optimising any single one.

9.3 Sensitivity to the central design choice

The single most consequential choice in the paper is the **co-designed implement library**. Without the co-design framing — that is, with conventional tractor implements substituted in place of the codesigned ones — the analysis weakens in three concrete places:

1. The 2kW **PMSM** is undersized (peak draft $\times$ tractor speed requires ≈ 5 kW), and the inverter, gearbox, and cooling stack scale up with it.
2. The energy-per-decare claim weakens by $\approx 2\times$ (because the draft drops by $\approx 2\times$ under co-design via the strip-width and v^2 levers), which directly weakens the off-grid PV+battery sizing.
3. The 9-auger Anchor loses its worst-case-soil margin: a conventional chisel/sweep plow at 14–15 kN P50 would need 41–45 augers on the worst-case Khand (2024) [4] loose-sand bound, although the same load fits in 9 augers on the Magnum (2024) [5] medium-dense nominal.

The first two bullets are the load-bearing ones — they hold for *any* soil regime — and the economics still work even in this degraded scenario (**NPV** stays positive at 25 ha/yr), but the off-grid story and the compaction story are weaker. The co-designed framing is therefore not a marketing convenience; it is the load-bearing assumption of the paper for energy and motor cost reasons that are independent of soil conditions, with the worst-case-soil anchor envelope as a bonus third leg, and we make this explicit. **Section 8.1** quantifies this by propagating a partial co-design shortfall through the full pipeline: the energy and off-grid claims degrade smoothly while the off-grid replacement-frame **NPV** stays essentially invariant, confirming which legs are sensitive.

9.4 What the paper does not claim

We do not claim:

- That CableTract is a finished product or has been prototyped.
- That cable-driven agriculture is a new idea (US 8,763,714 B2 [1] predates this work).
- That CableTract beats every competitor on every metric (EcoRobotix is more efficient per hectare on spraying-only; Naio is cheaper).
- That off-grid operation is universal (it is climate-conditional).
- That the $73\times$ contact-energy reduction translates linearly into a yield increase (the soil-yield link depends on crop, climate, rotation, and irrigation).

We do claim, with every number reproducible from the open analytical pipeline:

- A $\sim 4\times$ reduction in useful (drawbar) energy per hectare versus a diesel tractor ($\sim 13\times$ on a primary-fuel basis, most of the extra factor being generic ICE-to-electric conversion efficiency).
- A 97–98 % reduction in compacted area and a $\approx 73\times$ reduction in contact-energy index.
- Off-grid feasibility in solar-rich climates with 15 m^2 **PV** + 9 kWh battery.
- **NPV** positive across 1–100 ha at 5/8/12 % discount rates, with discounted payback < 1 yr at ≥ 25 ha.
- Lifecycle CO_2 $2.2\times$ lower than diesel, $1.6\times$ lower than a Monarch-class electric tractor.
- An NPV-positive operating envelope in 100 % of 3 600 (annual **GHI** $\times$ farm size) cells, with discounted payback never exceeding 3.06 yr.

10 Limitations

This section separates two categories of limitation. **Section 10.1** lists the *analytical* limitations of the present model — what the framework does not include. **Section 10.2** lists the *engineering* risks that a prototype phase would have to resolve regardless of how comprehensive the analytical framework becomes; in the authors’ candid view these are the dominant near-term feasibility questions for the architecture and they are not closed by additional simulation.

10.1 Analytical limitations of the present model

1. **No prototype.** The strongest objection. Every number in the paper is from first-principles calculation; none has been validated against a working machine. We have tried to mitigate this by using [ASABE D497.7 \[2\]](#) (an experimentally calibrated standard), the conservative end of the helical-pile lateral capacity literature (Khand et al. 2024 [4] for loose-sand free-head bounds, Magnum Piering 2024 [5] design tables and the CHANCE Technical Design Manual [6] for the medium-dense fixed-head reference), Keller and Lamandé (2010) [15] (experimentally measured tractor contact pressures), open published [LCA inventories \[16, 17, 18, 19, 20, 21\]](#), and a Sobol global sensitivity analysis [23, 24, 25] to bound the uncertainty.
2. **Reduced-order, not full multiphysics, control simulation.** [Section 8.3](#) now adds a closed-loop depth-control co-simulation (a lumped-mass cable plus a unilateral gauge-wheel contact and a bandwidth-limited down-pressure actuator under PID and MPC), and [Sections 8.2](#) and [8.4](#) add a nonlinear p - y anchor model and a transient cable-recoil model. These are reduced-order, single-axis models validated against analytic limits, not a coupled full-multiphysics simulation of cable tension, three-dimensional anchor stability, and depth control together; that coupling, and its validation against hardware, remains for a prototype phase. [Section 10.2](#) continues to treat the depth-control loop as the dominant unresolved risk, now bounded rather than unexamined.
3. **Static contact-pressure compaction model.** No dynamic load transfer, no tyre deflection, no multi-pass cumulative deformation, no soil-yield link, and the carriage load excludes the vertical reaction of cable tension and tool downforce. The model is intentionally simple; a richer model would add compaction to the tractor side (multi-pass deformation) but also to the carriage side (repeated narrow-strip trafficking, added carriage downforce), so we do not assume it can only favour CableTract.
4. **Polygon corpus is small and curated.** 50 fields, hand-tuned, biased toward small awkward shapes. A real deployment would need a corpus pulled from EuroCrops or [\(US\) Department of Agriculture, National Agricultural Statistics Service \(USDA NASS\)](#) at scale.
5. **PV harvest model in the operating-envelope sweep is a one-parameter linear fit.** The full hourly TMY pipeline lives in [Section 5.7](#) and was used for [Figs. 10 to 12](#), but the (annual $\text{GHI} \times \text{farm size}$) sweep in [Fig. 24](#) uses a calibrated linear approximation for tractability. The $\alpha = 0.169$ value was fitted as the median across the 6 bundled sites.
6. **The codesigned implement library is a paper exercise.** The 10 implements are sized using the [ASABE D497.7](#) equation but have not been physically built. The draft reductions are predictions, not measurements.
7. **The competitor comparison uses public-domain manufacturer claims.** Some of these may not reproduce in the field. The source of every number is flagged in the bundled competitor table.

10.2 Open mechanical risks

The seven items above are the analytical limitations of the present model: they describe what the framework does not include. This subsection separates out a different category — **the engineering risks that a prototype phase would actually have to resolve**, regardless of how comprehensive the analytical framework becomes. We list them explicitly so a follow-on team can attack them in priority order rather than re-deriving the list from the manuscript.

Closed-loop tilling depth control. Tilling depth at the carriage is governed by the geometric coupling between cable sag, cable elastic stretch, carriage horizontal position, implement reaction torque, and the time-varying soil contact force. None of these are independent variables, and none are simulated under closed-loop control in the present paper. [Sections 5.4.2](#) and [5.4.3](#) bound the

static sag-vs-tension tradeoff under nominal loads, but do not address the dynamic response to a transient draft spike — a buried stone, a hardpan layer, a moisture gradient. Unlike a tractor with a 3-point hitch and several tonnes of vehicle inertia to absorb the spike, the cable carriage has neither inertia nor a stiff mechanical reference frame; the response is mediated entirely by cable elasticity and the winch motor’s torque-control bandwidth. Whether the resulting tilling quality meets agronomic acceptance criteria is, in the authors’ view, the single largest unresolved technical risk in the architecture. It is also one of the cheapest to test: a quarter-scale bench rig on a 5 m span in a soil bin is sufficient to bound it before any full-scale prototype is built. The closed-loop co-simulation added in [Section 8.3](#) now bounds this risk analytically: it confirms the cable is far too soft to set depth (so a gauge wheel must), shows that a tuned controller holds the ± 2 cm tolerance through stone, hardpan, and moisture disturbances, and isolates the residual failure mode — transient gauge-wheel lift-off during a sharp draft spike — which is removed by carrying a steady down-pressure reserve of ≈ 1.75 kN at a ≥ 5 Hz actuator bandwidth. The bench rig remains the right validation step; what changes is that it now has a quantitative target to confirm rather than an open question to explore.

Per-strip setup time as a design target rather than a measurement. The model uses $t_{\text{setup}} = 60$ s as the per-round overhead ([Table 3](#)). This is a *design target*, not a field measurement, and the architecture in [Section 3.1](#) relies on the only configuration that makes it plausible: nine *parallel* auger drives on the Anchor (one BLDC per auger, all nine cycling concurrently) and four parallel auger drives on the Main Unit. With parallel augering, the per-round cycle decomposes notionally into ~ 30 s for the simultaneous nine-auger insert/retract pair, ~ 5 s for the lateral Anchor step on its own wheels, ~ 5 s for cable slack take-up, ~ 5 s for GNSS re-survey of the new strip line, and ~ 5 s for re-tensioning — about 50 s of nominal work inside the 60 s budget, with roughly 10 s of margin. The budget is tight but internally consistent with the [Section 3.1 BoM](#); the time-budget finding of [Section 5.9.5](#) (operation dominates setup by $\sim 10\times$) holds under it. Three residual concerns survive and should be measured by a prototype campaign before this number is treated as solved. (i) The single-auger insert/retract time at this duty cycle is unmeasured in the public literature; if it turns out to be significantly longer than 30 s on representative soils, the entire 60 s budget shifts. (ii) The margin inside the budget is small enough that any non-augering surprise — a tangled cable, a carriage hookup failure, a stale GNSS fix — consumes the slack on the round it occurs. (iii) Auger insertion time on hard-pan or stony soils is non-linear in cone index in ways the present model does not capture. The Sobol sweep ([Section 5.11.3](#)) brackets setup time over 30–180 s, which covers the realistic uncertainty around the parallel-drive nominal but does not extend to the failure mode where parallel augering itself is slower than assumed. The CableTract+ variant in [Appendix A](#) eliminates the per-strip Anchor reset entirely and is, in part, a structural insurance policy against this concern rather than an independent throughput play.

Anchor self-fatigue of its own anchoring soil. The Anchor’s lateral capacity envelope ([Section 5.4.4](#)) treats per-auger lateral capacity as a property of soil texture and density alone. In normal operation, however, each Anchor station drives and retracts nine augers tens to hundreds of times across a cropping season as the Anchor walks the headland. The cumulative effect is that the soil under each station is progressively loosened by the system’s own operation: a soil that began as medium-dense (and therefore sat comfortably inside the Magnum 2024 [\[5\]](#) envelope at $5\text{--}10\times$ margin) drifts toward the loose-sand regime (Khand 2024 [\[4\]](#) envelope at $\sim 1\times$ margin) as the season progresses. The 9-auger design margin erodes *from inside*, not from changes in field conditions. Quantifying the rate at which this happens requires either an empirical campaign on a fixed Anchor station or a finite-element model of repeated helical pile insertion and withdrawal; neither is in the present paper, and the architecture’s worst-case soil envelope should be read with this caveat in mind.

Helical pile retraction wear. Helical screw piles in the construction industry are designed and qualified for one-shot installation with permanent service. CableTract drives and retracts each auger many cycles per day, which is a regime for which no fatigue, helix wear, shaft buckling, or soil-disturbance data exists in the public literature the authors could find. The first-order risks are blunting of the helix leading edge (which would slow each subsequent insertion and increase the auger drive’s torque demand), shaft fatigue at the drive coupling, and progressive misalignment of the helix from the shaft axis. The framing is that this is an unmeasured failure mode rather than necessarily a small one — there is no published basis to estimate the mean time between auger replacements, and the bill of materials in [Section 5.8](#) does not include an auger replacement schedule. A first prototype campaign should treat auger wear as a primary measurement endpoint alongside throughput and energy.

Historical cable ploughing as an unaddressed precedent. The architecture has a precedent the present paper has not yet fully engaged with. Steam-driven cable ploughing was a real and widely deployed technology in the second half of the 19th century, particularly on heavy clay soils in the United Kingdom and parts of continental Europe where horse teams bogged. Two stationary steam engines at opposite headlands pulled a two-bottom plough back and forth on a steel cable (the Fowler and Howard cable-ploughing systems). The system coexisted with horse and traction-engine ploughing for several decades and then disappeared, comprehensively, by the early 20th century. The reasons usually given for its decline — which we cite here from secondary recollection rather than a primary historical source we have verified, and flag as such — are per-station crew overhead, the time cost of repositioning the engines after each strip, the management burden of the cable itself, and the inability to turn at the headland and resume continuously. The present manuscript automates the per-station crew problem (the Main Unit and Anchor are autonomous) but does not yet solve the repositioning, cable-management, or continuous-headland-turn problems that killed the steam-era version. A serious treatment of this lineage — what cable ploughing did, why it failed, what is mechanically and operationally different now — belongs in any prototype-stage publication. We flag its absence here as a deliberate scope decision and a known weakness.

Cable safety surface. A 50 m taut steel or Dyneema cable at 1.5–4 kN pretension running across an open field at 0.8 m above the ground is a safety surface the paper does not address. Snap-back energy from a cable failure is non-trivial at these tensions, and a deployed system would need cable-condition monitoring, runaway-detection logic, exclusion zones, and certification under the relevant agricultural-machinery regulatory framework. None of this is in scope here, but it would be a precondition for any field deployment and would add both [BoM](#) cost and certification process cost. [Section 8.4](#) now quantifies the dominant terms: snap-back recoil velocities (counter-intuitively higher for the lighter synthetic rope) set the required full-span exclusion zone; the bending-fatigue life is shown to be UV/abrasion-limited at ≈ 8 yr rather than fatigue-limited, which supplies the previously-missing cable-replacement cost and life-cycle line ([Section 5.8](#)); and an aeroelastic check flags vortex-induced vibration (requiring dampers) and an iced-galloping constraint (de-tension in icing). These bound the safety surface analytically but do not substitute for the monitoring, certification, and field validation a deployment still requires.

The unifying observation is that the strongest near-term feasibility questions for CableTract are **mechanical and operational**, not financial or environmental. The paper’s NPV, life-cycle CO₂, and energy-per-hectare numbers are robust within their stated framework; the open question is whether the framework’s mechanical assumptions survive contact with a real prototype season. We make this distinction explicit so a reviewer can evaluate the analytical contribution on its own merits while understanding which questions a prototype-stage follow-up would actually need to answer.

11 Conclusion

This paper presents a prototype-free, fully reproducible feasibility envelope for **CableTract**, a co-designed cable-driven field robot consisting of a stationary **MU**, a lighter Anchor anchored by helical screw piles, and a lightweight implement carriage that rolls along a tensioned cable between them. The central design lever is the **co-designed implement library** — 10 implements sized for the cable architecture rather than borrowed from conventional tractor inventories — which reduces median draft by $\approx 63\%$ and brings the entire codesigned library inside the 9-auger Anchor envelope on the medium-dense soil reference, with the lighter half of the library also fitting inside the worst-case loose-sand envelope.

The codesigned reference achieves:

- **889 Wh/decare** ($\sim 4\times$ less useful/drawbar energy, $\sim 13\times$ less primary fuel energy, than a diesel tractor; 921 Wh/decare conservative flat-field bound without regen).
- **97–98 % reduction in compacted area** and $\approx 73\times$ **reduction in contact-energy index** across rectangle, L-shape, and irregular-concave field classes.
- **Off-grid throughput of 10–14 decares/day** at six bundled sites under the climate-conditional framing of [Section 5.7.4](#).
- **Capex parity with a used diesel tractor** (35 870 EUR vs 35 000 EUR): for a buyer *replacing* a tractor, NPV **3 575 EUR** at 25 ha/yr (8 % discount, 1.30 yr payback on the 870 EUR increment), positive across (1–100 ha) at all three discount rates; a buyer financing the whole machine instead sees a negative NPV until ~ 240 ha/yr.
- **Lifecycle CO₂ of 14.6 kg/(ha yr)** versus 32.5 for diesel — a $2.2\times$ improvement that does not depend on grid decarbonisation.

The operating envelope of [Fig. 24](#) shows that, for a replacing buyer, CableTract is **NPV-positive** in 100 % of the 3 600 cells of the (annual **GHI** $\times$ farm size) sweep, with discounted payback never exceeding 3.06 yr ($\approx 22\%$ of cells under the additive frame). The binding *physical* constraint is the annual energy balance for large operated areas in low-**GHI** climates, not financial viability: CableTract’s operating envelope is more a *climate envelope* than a *financial* one.

The right next step is a **Main Unit prototype** with a 2 kW **PMSM**, a 9 kWh pack, a 9-auger Anchor, and the codesigned narrow ripper as the first implement, deployed on a 25 ha rectangular field at a Mediterranean site (Konya or Palencia) for one full cropping season. The data this prototype would generate — actual draft under load, actual cable tension regulation, actual anchor stability, actual energy harvest — would close the largest gaps in the present paper. Until then, the analytical envelope above is the best defensible feasibility case for the architecture.

A Architectural variants considered

The architectural family has several siblings worth naming. Regenerative braking on the return leg is part of the default baseline ([Section 5.4.1](#)). Two configurations are quantitatively compared against that baseline in [Section 5.12](#) and [Fig. 22](#) — CableTract+ and the unidirectional (no-regen) drivetrain; two more are implemented in `cabletract/variants.py` but excluded from the comparison because the time-budget finding of [Section 5.9.5](#) shows they act on the wrong slice (setup is only 9 % of the daily time budget); two are deferred to future work.

Table 18: Architectural variants and their modelling status. “Compared” means the variant is implemented as a parameter transformation on the codesigned reference, evaluated in [Section 5.12](#), and shown alongside the baseline in [Fig. 22](#).

Variant	Description	Status
Regenerative return leg	Four-quadrant motor operation recovering energy on the unloaded carriage return / downhill slope.	Default (baseline)
CableTract+	Four corner Main Units pulling the carriage with two simultaneously-active cables — eliminates the Anchor and the per-strip alignment overhead.	Compared
Unidirectional (no regen)	The baseline with the regen drive removed; one-way drivetrain, 300 EUR cheaper, $\approx 3.6\%$ higher flat-field energy.	Compared
Circular pulley	Main Unit’s output pulley swings on a vertical pin so the cable can leave the drum at angles up to $\pm 25^\circ$, allowing the MU to stay put while the Anchor steps laterally.	Excluded (setup-side)
Drone-assisted alignment	A small quadcopter drops GPS markers between fields so re-alignment takes seconds.	Excluded (setup-side)
Liquid-hose extension	The cable doubles as a liquid feed for spraying or fertigation.	Deferred
In-place rotary mode	The carriage spins around the MU on a single tether for circular operations such as orchard mowing.	Deferred

B List of acronyms

4WD Four-Wheel Drive

API Application Programming Interface

ASABE American Society of Agricultural and Biological Engineers

BLDC Brushless DC motor

BloombergNEF Bloomberg New Energy Finance

BMS Battery Management System

BoM Bill of Materials

BoS Balance of System (PV cabling, mounting, inverter)

CAPEX Capital Expenditure

DEFRA (UK) Department for Environment, Food & Rural Affairs

EEA European Environment Agency

FEM Finite Element Method

GHI Global Horizontal Irradiance

GNSS Global Navigation Satellite System

GPS Global Positioning System

ICE Internal Combustion Engine

ICE v3 Inventory of Carbon and Energy, version 3 (Hammond & Jones)

INMET (Brazil) National Institute of Meteorology

IVL IVL Swedish Environmental Research Institute

IWRC Independent Wire Rope Core

LCA Life-Cycle Assessment

MU Main Unit (winch + motor + battery + harvester module)

NIWE (India) National Institute of Wind Energy

NPV Net Present Value

NREL (US) National Renewable Energy Laboratory

NSRDB (NREL) National Solar Radiation Database

P10 10th percentile of a sampled distribution

P50 50th (median) percentile of a sampled distribution

P90 90th percentile of a sampled distribution

PMSM Permanent-Magnet Synchronous Motor

PV Photovoltaic

PVGIS (EU JRC) Photovoltaic Geographical Information System

RQ Research Question

S1 Sobol first-order sensitivity index

SALib Sensitivity Analysis Library (Python)

SARAH2 Surface Solar Radiation Data Set — Heliosat, version 2

SoC State of Charge

ST Sobol total-order sensitivity index

SWERA Solar and Wind Energy Resource Assessment

TMY Typical Meteorological Year

USDA NASS (US) Department of Agriculture, National Agricultural Statistics Service

VAWT Vertical-Axis Wind Turbine

CRedit authorship contribution statement

Özgür Yılmaz: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing — original draft, Writing — review & editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The complete analytical code, all bundled input data (TMY summaries, ASABE D497 coefficients, helical-pile capacities, cable mechanical properties, BoM CO₂ intensities, and the field-polygon corpus), and every figure and result table in this manuscript can be regenerated from a clean checkout by running the phase scripts (run_phase1–run_phase12); per-figure CSV companions are written to tables/. The repository is openly available at <https://github.com/yilmazozgur/cabletract> and is archived at Zenodo (DOI to be assigned on acceptance).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used a large language model (Anthropic Claude) to assist with implementing and structuring the analytical code, drafting and editing prose, and organising the presentation of results. After using this tool, the author reviewed and verified all output against the executed code, edited it as needed, and takes full responsibility for the content of the published article.

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